

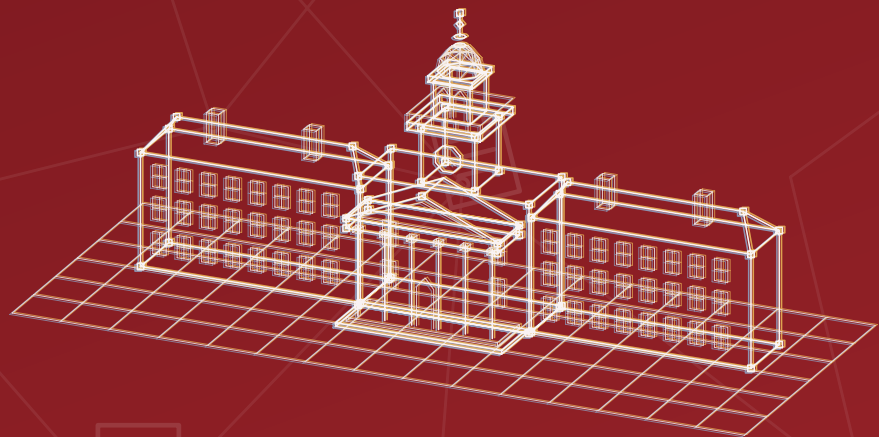


Est. 2020 as a terminal script for ResNet Support Representatives.  
Rebuilt 2026 as a monitoring and analytics platform.

# BSU Student Printing Ops

*The print-station monitor, documented end to end*

How Bridgewater State's 31 Wepa print stations are watched, measured, explained and improved, told through the full CRISP-DM lifecycle: from the business problem to the deployed product, with a guide for everyone who uses it.



31 stations · 16 ResNet (green) · 14 Student Computer Labs (gold) · 1 Satellite Campus (white)

DOCUMENT  
SPO-DOC-001

PRODUCT VERSION  
1.7.0 · October 2026

METHOD  
CRISP-DM, six phases

STATUS  
Released · living document

Designed, built and documented by **Nicholas Freberg**, Bridgewater State University Class of 2020 · Former BSU ResNet Support Representative (2018–2020).

An independent portfolio project. It is not an official publication of Bridgewater State University or Wepa, and it does not represent either. Wepa's status page is read for internal operations only; its data is not republished.

# About this document

One document for everyone who touches the product: the student whose printer jammed at 11 PM, the student worker on shift, the supervisor, IT leadership, and the engineer or data scientist who will maintain it next.

| Document control |                                                                                              |
|------------------|----------------------------------------------------------------------------------------------|
| FIELD            | VALUE                                                                                        |
| Title            | BSU Student Printing Ops: product documentation (CRISP-DM lifecycle edition)                 |
| Identifier       | SPO-DOC-001 , kept with the code in docs/product/                                            |
| Describes        | Product version 1.7.0 (see the change log in the app and in CHANGELOG.md )                   |
| Author           | Nicholas Freberg                                                                             |
| Classification   | Internal operations. Contains no personal data; screenshots use the synthetic demo data set. |
| Dates            | Written ISO 8601 (YYYY-MM-DD); times are Eastern (America/New_York), as in the app           |
| Review           | Revise with every minor release; the change log names what changed                           |

## How to read it

The chapters follow CRISP-DM, the Cross-Industry Standard Process for Data Mining, because the product was built that way: understand the problem, understand the data, prepare it, model it, evaluate the models honestly, then deploy and keep watching. Chapters 7 to 9 step outside that cycle for process models, the user guide and standards.

Every chapter opens with **In short**, a plain-language summary anyone can read. Boxes marked with an audience give a little extra for that reader. Badges at the top of each chapter say who it is mainly for.

## Students

Read 8.1 Finding a printer that works. Two minutes.

## Student staff

The summary, then the user guide, especially Overview, Rounds and Investigations.

## Supervisors

Business understanding, the user guide, and process models.

## IT leadership

The summary, evaluation and deployment, then the standards.

## Engineers

Data preparation, the data dictionary, architecture and the appendices.

## Data scientists

Data understanding through evaluation; the model inventory is in chapter 4.

# Contents

## 0 Executive summary

What it is, why it exists, what it does

## 1 Business understanding

Problem, stakeholders, goals, journeys, requirements

## 2 Data understanding

Sources, cadence, quirks, quality

## 3 Data preparation

Pipeline, cleaning rules, incidents, data dictionary

## 4 Modeling

Metrics, statistics, machine learning, AI analyst, decision tables

## 5 Evaluation

Gates, testing, model scoring, limits

## 6 Deployment

Architecture, Azure, availability, security, updates

## 7 Process models

BPMN, CMMN, DMN and Six Sigma views

## 8 User guide

Every page, task by task, for every role

## 9 Standards and recommendations

ISO, ACM, ASA, OMG, Six Sigma, DASCA, data ecology

## A Appendices

Configuration, commands, glossary, sources, credits

# A minute-by-minute record of every campus printer

Everyone

IT leadership

Wepa publishes a status page for BSU's print stations that refreshes every minute and only ever shows *now*. BSU Student Printing Ops reads it once a minute, keeps every reading, and turns the history into answers: what is broken, for how long, why, when nobody was there to fix it, what to restock, and where a student should walk instead.

**Mission.** Keep every BSU print station ready when students need it, by measuring service the way students experience it and acting on what the data shows.

| MEASURE                                                                                                           | ACT                                                                                                                            | IMPROVE                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Every station, every minute: availability, outages, causes, supplies and usage, computed the same way every time. | Alerts, prioritized rounds, supply forecasts and inventory turn findings into the next visit, the next order and the next fix. | Investigations, evidence for Wepa, placement reviews and reporting close the loop, so service gets measurably better each term. |

## WHAT IT WATCHES

31 stations: 16 ResNet printers in and around the residence halls, 14 in Student Computer Labs, 1 at a satellite campus.

## HOW OFTEN

One reading every 60 seconds, every reading kept. Failed reads are kept too, so gaps are visible rather than guessed.

## WHO IT SERVES

Students, ResNet and IT Service Center staff, supervisors and IT leadership, each with their own view.

## HOW IT IS BUILT

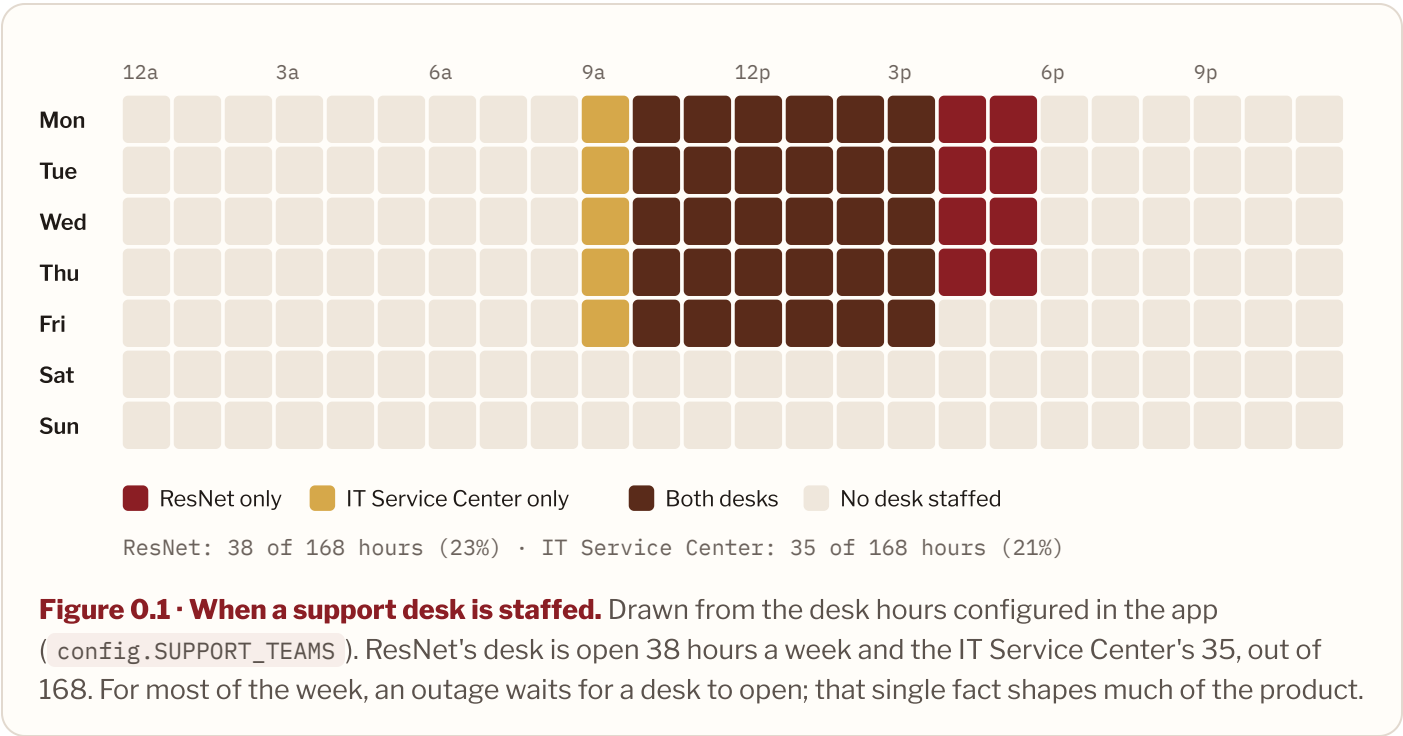
Python, pandas and Dash; about 24,000 lines of application code and more than 200 automated tests; deployed on Azure Container Apps.

# What it does

- **Now:** every station's status in one vocabulary (Operational, Degraded, Out of service, No signal), a ranked work queue, and a walking or van route to everything that needs a visit.
- **Then:** availability, time to repair, time between failures, faults by cause and hour, supplies and parts, all with stated definitions and minimum-evidence rules.
- **Why:** plain-language stories, click-to-explain charts, statistical tests, and an optional AI analyst that may only state figures it looked up.
- **Next:** supply reorder points, end-of-life forecasts for parts, and an outage-risk model that is only shown when it beats a simple baseline.
- **Accountability:** permanent reference numbers for every outage, governed investigations with a tamper-evident trail for Wepa cases, named staff accounts and an audit log.
- **Supplies and keys:** an inventory of parts and paper from delivery to the printer each unit went into, invoices read automatically and approved by a second person, counts that open an investigation when stock is missing, weekly paper checks, and a log of who holds a Wepa kiosk key.
- **Self-maintenance:** daily vulnerability checks ranked by real-world exploitation, reviewed update packages, backup collectors and a self-check.

# Who uses it, and where to start

| AUDIENCE                         | START HERE                                         | WHAT THEY GET                                                                                                 |
|----------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| IT leadership                    | Executive summary · IT Outcomes                    | Service levels for the month and year, what changed, and a print-ready annual-report feature                  |
| Managers and supervisors         | Overview · Analytics · Investigations · Inventory  | What needs attention, the station report card, root-cause work with an audit trail, and supply accountability |
| Data engineers and statisticians | Analytics › Forecasts & statistics · System health | Stated definitions, sample sizes, confidence intervals, model back-tests and data-quality checks              |
| Student workers                  | Overview · Rounds · Inventory › Record             | The next printer to visit, the route, and a place to record deliveries, moves and counts                      |
| Students                         | The student status page                            | Which printers work right now, and the nearest working one they can walk into                                 |



# Business understanding

Supervisors

IT leadership

Everyone

## IN SHORT

Students need to print, often late and close to a deadline. When a printer breaks, the student finds out at the printer and staff find out later. The product exists to shorten that gap, to show where the gaps come from, and to give the university evidence to act on, without adding work for student staff.

## 1.1 Background

Bridgewater State University offers Wepa print stations across campus: in residence-hall areas (supported by ResNet, the residential network team based in East Campus Commons), and in labs, the library, the student union and a satellite campus (supported by the IT Service Center in Maxwell Library). Students pay per page through Wepa and print papers, forms, tickets and boarding passes.

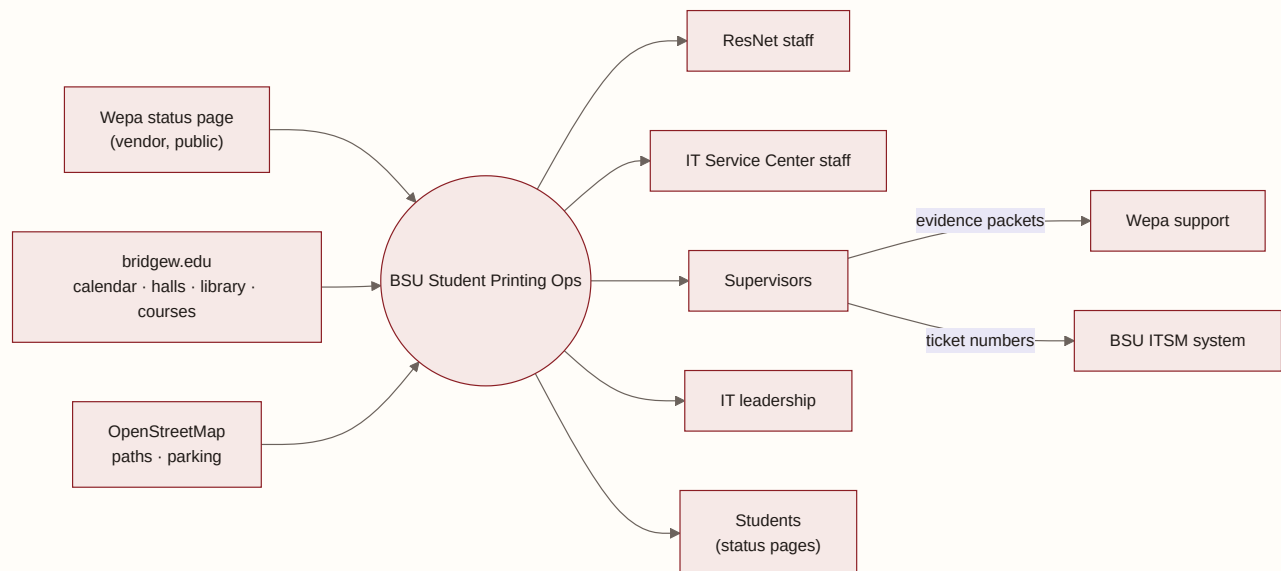
Wepa publishes a public status page listing every station with a color (green, yellow, red), alert codes, printer text and consumable levels. It refreshes every minute and shows only the present. The original version of this product (2020) was a terminal script for ResNet Support Representatives: choose your assigned area and it beeped when a printer there went down.

## 1.2 Problem statement

Nobody could answer basic operational questions with evidence: which printers break most; how long outages last and why; whether problems wait overnight because the desks are closed; what supplies will run out next; and where a student should go when their printer is down. Without history, recurring hardware faults looked like bad luck, and conversations with the vendor relied on memory.

## 1.3 Stakeholders

| STAKEHOLDER                    | NEEDS                                                 | USES                                                   |
|--------------------------------|-------------------------------------------------------|--------------------------------------------------------|
| Students                       | A working printer nearby, especially after hours      | Student status pages (QR signs), indirectly everything |
| ResNet Support Representatives | Know what is broken, where to go first, what to carry | Overview, Rounds, station pages, Investigations        |
| IT Service Center staff        | The same, for labs and public spaces                  | Overview, Rounds, Stations                             |
| Supervisors (ResNet, ITSC)     | Coverage, staffing, recurring faults, vendor evidence | Analytics, Planning, Investigations, Insights          |
| IT leadership                  | Service level, trends, outcomes to report             | Executive summary, IT Outcomes, exports                |
| Residence Life                 | Printing in the halls students live in                | Reports from supervisors                               |
| Wepa (vendor)                  | Clear, dated evidence when hardware fails repeatedly  | Investigation evidence packets (PDF)                   |



**Figure 1.1 · Context.** What flows into the product and who it serves. The ITSM system and Wepa support are referenced by number only; nothing is synchronized with them.

## RESPONSIBILITIES (PROPOSED RACI)

| ACTIVITY                            | STUDENT STAFF | SUPERVISOR | IT LEADERSHIP | ADMINISTRATOR | WEPA    |
|-------------------------------------|---------------|------------|---------------|---------------|---------|
| Respond to an outage                | R             | A          | I             | –             | C       |
| Open and work an investigation      | R             | A          | I             | –             | C       |
| Review and close an investigation   | –             | R/A        | I             | –             | –       |
| Escalate a hardware pattern to Wepa | C             | R/A        | I             | –             | R (fix) |
| Monthly and annual reporting        | –             | R          | A             | –             | –       |
| Accounts, deploys, security updates | –             | C          | I             | R/A           | –       |

R responsible · A accountable · C consulted · I informed. This is a recommendation for how the roles map onto the product; it is not a published BSU policy.

## 1.4 Objectives and success criteria

CRISP-DM separates business objectives from the data-mining goals that serve them. Success is measured by the product's own figures once enough live history exists.

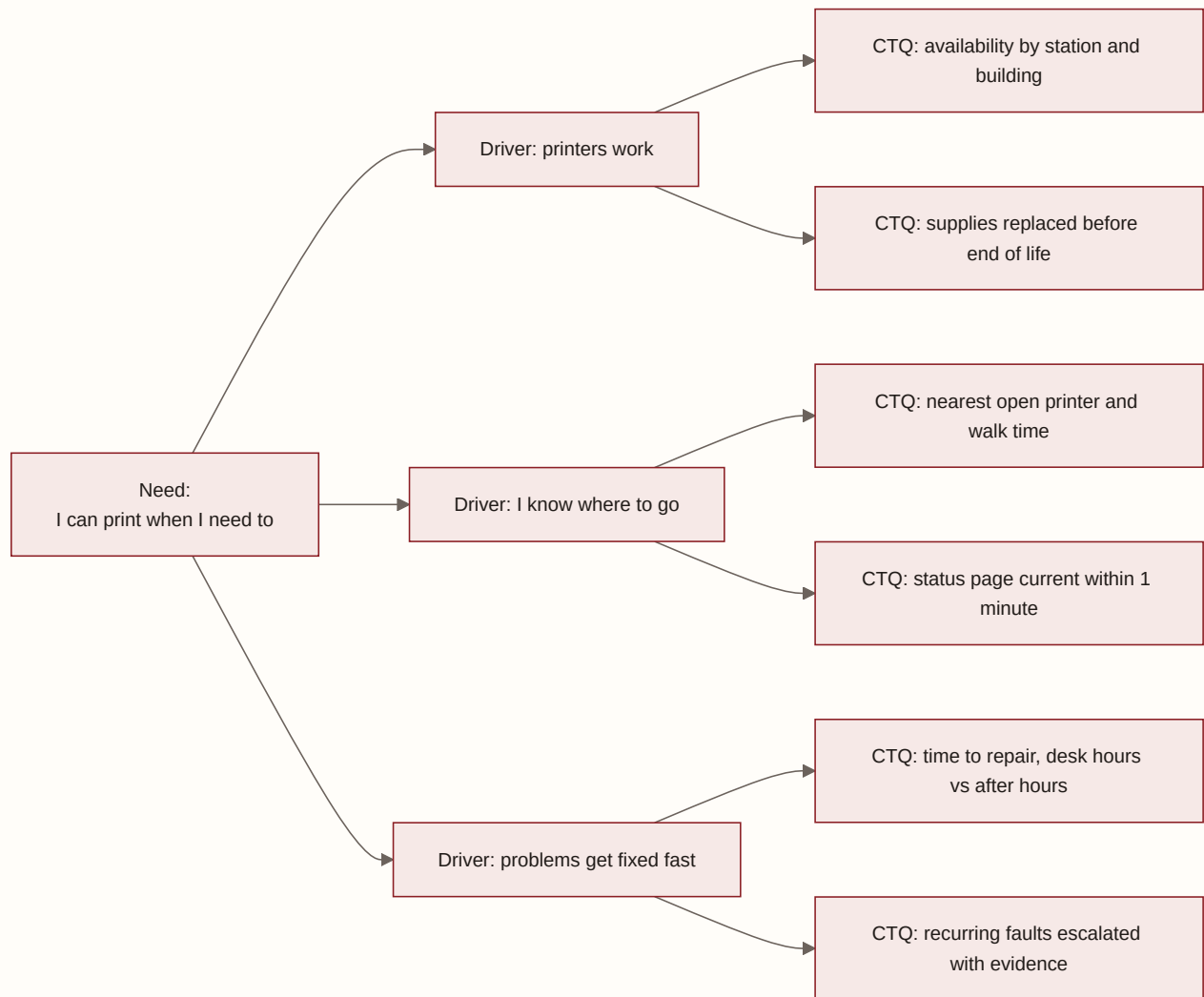
| BUSINESS OBJECTIVE                  | DATA GOAL                                                                          | SUCCESS CRITERION                                              |
|-------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Fix printers sooner                 | Detect each outage within a minute; rank the queue by impact                       | Median time to repair falls term over term                     |
| Fewer outages that wait overnight   | Split downtime into desk hours and after hours per team                            | Share of downtime that starts after hours is known and tracked |
| Evidence for the vendor             | Permanent reference numbers; recurring-fault detector; hash-chained investigations | Every Wepa case backed by a dated evidence packet              |
| Supplies before they run out        | Usage that survives part replacements; end-of-life forecasts; reorder points       | No station out of a consumable that was forecast               |
| Point students to a working printer | Walking-time network between buildings; access rules                               | Every printer has a named backup and walking time              |
| Trustworthy numbers                 | Data-quality score; evidence gates; no imputation                                  | Data quality 95 or above; every figure states its basis        |

## 1.5 SIPOC

A Six Sigma SIPOC places the product in the service it supports: suppliers, inputs, process, outputs, customers.

| SUPPLIERS                                                                          | INPUTS                                                                                                                            | PROCESS                                                                        | OUTPUTS                                                                                                 | CUSTOMERS                                                            |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Wepa<br>BSU web pages<br>OpenStreetMap<br>Open-Meteo<br>Vulnerability<br>databases | Station status<br>every minute<br>Calendar, halls,<br>library hours,<br>courses<br>Paths, parking<br>Hourly weather<br>Advisories | Collect → store →<br>detect incidents →<br>measure → explain →<br>act → review | Live status and<br>work queue<br>Routes<br>KPIs and reports<br>Investigations<br>and evidence<br>Alerts | Students<br>Student<br>staff<br>Supervisors<br>IT leadership<br>Wepa |

## 1.6 Voice of the customer, critical to quality



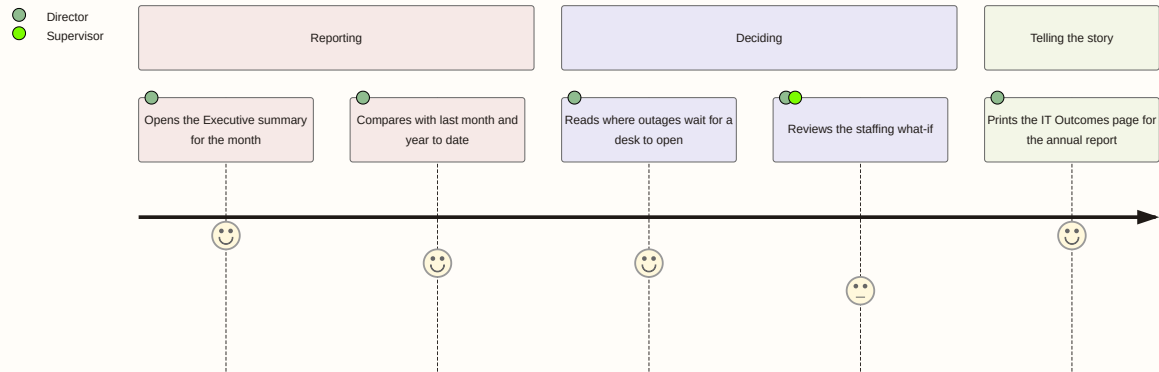
**Figure 1.2 • CTQ tree.** From the student's need to measurable characteristics the product reports.

## 1.7 Customer journeys

Three journeys, scored 1 (frustrating) to 5 (smooth). The scores are design assessments of the experience with the product in place, not survey results.



## IT leadership at month end



**Figure 1.5 · Leadership journey.** The same numbers staff work from, summarized, with their definitions attached.

## 1.8 Constraints, assumptions and risks

| TYPE       | ITEM                                                        | RESPONSE                                                                                                          |
|------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Constraint | Wepa's terms allow personal, non-commercial use of its site | Data is used for internal operations; not republished; written permission from Wepa and BSU before any public use |
| Constraint | The status page can change format without notice            | Columns are found by header text; a mismatch fails loudly and is logged                                           |
| Constraint | No access to Wepa's internal systems or job logs            | Usage is inferred from consumable levels; stated as such                                                          |
| Assumption | The row color is Wepa's own verdict of severity             | Used directly; codes only add names and fix categories                                                            |
| Assumption | Desk hours in configuration are the dependable coverage     | Editable in one place; holidays from the academic calendar                                                        |
| Risk       | Monitoring gaps (network, restarts)                         | Gaps recorded and excluded, never imputed; backup collectors                                                      |
| Risk       | Over-reading thin evidence                                  | Minimum-evidence gates; sample sizes on every statistic                                                           |
| Risk       | AI stating something untrue                                 | Read-only tools, a fact check of every number, links and HTML stripped                                            |
| Risk       | Vulnerable dependencies                                     | Hash-pinned installs, daily scans, threat-intelligence ranking                                                    |

## 1.9 Requirements

Requirements are prioritized with MoSCoW and traced to the module that implements them. Status reflects version 1.7.0. Quality requirements are grouped by the ISO/IEC 25010 characteristic they serve.

## Functional requirements

| ID    | REQUIREMENT                                                                                       | PRIORITY | STATUS | WHERE                    |
|-------|---------------------------------------------------------------------------------------------------|----------|--------|--------------------------|
| FR-01 | Read every station's status once a minute and log every attempt, successful or not                | Must     | Done   | scrape.py , collector.py |
| FR-02 | Keep raw readings append-only; never edit them                                                    | Must     | Done   | store.py                 |
| FR-03 | Turn readings into outages, warnings, fault and paper-tray incidents with start, end and duration | Must     | Done   | events.py                |
| FR-04 | Count consumable usage correctly across part replacements                                         | Must     | Done   | consumables.py           |
| FR-05 | Show every station's current status in one vocabulary                                             | Must     | Done   | vocab.py , Overview      |
| FR-06 | Publish KPIs with written definitions and minimum-evidence gates                                  | Must     | Done   | metrics.py               |
| FR-07 | Give every outage a permanent reference number                                                    | Must     | Done   | refs.py                  |
| FR-08 | Named staff accounts managed centrally; sign-in on every page                                     | Must     | Done   | accounts.py , auth.py    |
| FR-09 | Monitor data quality and the collector itself                                                     | Must     | Done   | System page              |
| FR-10 | Clearly labelled synthetic demo mode                                                              | Must     | Done   | synth.py                 |
| FR-11 | Ranked work queue and a route planner (walking or van)                                            | Should   | Done   | ops.py , routing.py      |
| FR-12 | Nearest printer open to everyone, with walking time                                               | Should   | Done   | nearby.py                |
| FR-13 | Split downtime by desk hours; say when support is unavailable                                     | Should   | Done   | support.py               |
| FR-14 | Analytics: reliability, faults, supplies, usage, report card, planning, statistics                | Should   | Done   | Analytics page           |

|       |                                                                                                                                              |        |      |                           |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------|--------|------|---------------------------|
| FR-15 | Plain-language stories, Ask the data, click-to-explain                                                                                       | Should | Done | narrative.py , ask.py     |
| FR-16 | Governed investigations with a tamper-evident history and evidence packet                                                                    | Should | Done | investigations.py         |
| FR-17 | Activity feed and notifications                                                                                                              | Should | Done | activity.py               |
| FR-18 | Export any view as CSV, Excel, JSON or PDF                                                                                                   | Should | Done | export.py                 |
| FR-19 | Campus context: calendar, halls, library hours, class schedules                                                                              | Should | Done | campus.py , courses.py    |
| FR-20 | Daily vulnerability check, ranked by exploitation                                                                                            | Should | Done | vulns.py , threatintel.py |
| FR-21 | Self-check and OWASP Top 10 checklist                                                                                                        | Should | Done | selfcheck.py , owasp.py   |
| FR-22 | Executive summary and IT Outcomes report page                                                                                                | Could  | Done | insights.py               |
| FR-23 | Outage-risk model, shown only when it beats a baseline                                                                                       | Could  | Done | risk.py                   |
| FR-24 | AI analyst with read-only tools and a fact check (optional)                                                                                  | Could  | Done | ai.py , analyst.py        |
| FR-25 | Student status pages and printable QR signs (opt-in public)                                                                                  | Could  | Done | dashboard/public.py       |
| FR-26 | Update packages built and tested by a reviewed pipeline                                                                                      | Could  | Done | updates.py                |
| FR-27 | Backup collectors with a signed handshake                                                                                                    | Could  | Done | peer.py                   |
| FR-28 | Feature requests filed as GitHub issues                                                                                                      | Could  | Done | feedback.py               |
| FR-29 | Campus map and Google Earth export                                                                                                           | Could  | Done | geo.py                    |
| FR-31 | Inventory of parts and paper by location, with a trail of every unit, second-person approval of deliveries and investigations for shortfalls | Should | Done | inventory.py , ledger.py  |
|       |                                                                                                                                              |        |      |                           |

|       |                                                                                          |        |         |                           |
|-------|------------------------------------------------------------------------------------------|--------|---------|---------------------------|
| FR-32 | Read supply invoices (PDF or photo) into draft lines for approval                        | Could  | Done    | inventory.py , ai.py      |
| FR-33 | Wepa kiosk key log; a lost key opens an investigation (staff only, never shared with AI) | Should | Done    | keys.py                   |
| FR-34 | Grouped navigation, one filter bar and a phone tab bar                                   | Should | Done    | dashboard/app.py          |
| FR-30 | Single sign-on with BSU's Entra ID (MFA)                                                 | Should | Planned | Needs BSU IT registration |

### Quality requirements (ISO/IEC 25010 characteristics)

| ID     | CHARACTERISTIC                     | REQUIREMENT                                            | HOW IT IS MET                                                                                                        |
|--------|------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| NFR-01 | Reliability                        | The collector never stops on an error                  | Failures recorded as data; the loop continues                                                                        |
| NFR-02 | Reliability (availability)         | No collection gap from a deploy or a collector failure | Rolling deploys with a collector lock; backup takes over after 60 s of silence and stands down within one 15 s check |
| NFR-03 | Performance efficiency             | Pages reflect a new reading within about 15 s          | Incremental refresh; finished days read once (Parquet)                                                               |
| NFR-04 | Security (confidentiality)         | Every route needs a session; least privilege           | Tested by walking every route; roles; CSP                                                                            |
| NFR-05 | Security (integrity)               | Records provably unaltered                             | Append-only files; SHA-256 hash chain; hash-pinned installs                                                          |
| NFR-06 | Interaction capability (usability) | Readable by anyone                                     | One vocabulary; whole-unit durations; plain-language stories                                                         |
| NFR-07 | Accessibility                      | Keyboard, contrast and motion preferences respected    | Appearance menu; visible focus; reduced motion. A formal WCAG 2.2 audit is recommended                               |
| NFR-08 | Maintainability                    | Rules changeable in one place; changes verified        | <code>config.py</code> ; 190 automated tests; lint and scans in CI                                                   |
| NFR-09 | Flexibility (portability)          | Runs on a laptop, a VM or containers                   | Python 3.11/3.12; three documented deployments                                                                       |
| NFR-10 | Privacy                            | Collect the minimum about people                       | Visits shortened to network; no passwords logged; names kept out of GitHub issues                                    |
| NFR-11 | Statistical honesty                | Never present a guess as a measurement                 | No imputation; evidence gates; sample sizes; demo labelled                                                           |
| NFR-12 | Cost control                       | Optional AI cannot run away with spend                 | Calls per hour capped; answers cached                                                                                |

# Data understanding

Data scientists

Engineers

Supervisors

## IN SHORT

One web page is the heart of it: Wepa's status page, read every minute. Around it sit campus facts (the academic calendar, residence halls, library hours, class schedules), campus maps for walking routes, hourly weather, and security advisories. Each source has known quirks, and each is handled openly rather than smoothed over.

## 2.1 Sources

| SOURCE                                                                      | WHAT IT PROVIDES                                                                                                       | CADENCE                         | TERMS AND NOTES                                                                             |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------|
| Wepa status page<br><code>cs.wepanow.com/000BRIDGEW149.html</code>          | Per station: color (green, yellow, red), alert codes, printer text, toner, drum, belt and fuser levels, page timestamp | Every 60 s                      | Personal, non-commercial use under Wepa's terms; internal operations only                   |
| bridgew.edu                                                                 | Academic calendar, residence halls, library hours                                                                      | Calendar yearly, library weekly | Public university pages; committed copies kept as fallback                                  |
| BSU course search                                                           | Class meetings by building and room                                                                                    | Monthly                         | Public schedule search                                                                      |
| OpenStreetMap                                                               | Footpaths, roads and parking for routes and walking times                                                              | On demand                       | © OpenStreetMap contributors, ODbL                                                          |
| Open-Meteo                                                                  | Hourly Bridgewater weather (humidity)                                                                                  | Daily                           | CC BY 4.0 data                                                                              |
| OSV.dev, CISA, NIST NVD, Microsoft MSRC, FIRST EPSS, Exploit-DB, Metasploit | Known vulnerabilities in the app's own packages, and how urgent they are                                               | Daily (catalogs weekly)         | Only package names, versions and CVE numbers leave the server                               |
| Reference files (maintained by hand)                                        | Station to building, area, type and access; building coordinates; van parking                                          | When the campus changes         | <code>reference/stations.csv</code> , <code>buildings.csv</code> , <code>parking.csv</code> |

## 2.2 What a reading looks like

Each minute produces one row per station. A trimmed example, as stored:

| SCRAPE_TS            | SECTION | STATION_ID | DESCRIPTION         | ROW_STATUS | STATUS_CODE  |
|----------------------|---------|------------|---------------------|------------|--------------|
| 2026-10-02T23:41:00Z | ResNet  | 02132      | East Campus Commons | red        | paper_out_er |

Illustrative row in the stored format. Levels are percentages remaining; codes are normalized ( `Alert_paper_out_error` becomes `paper_out_error` ).

## 2.3 Known quirks

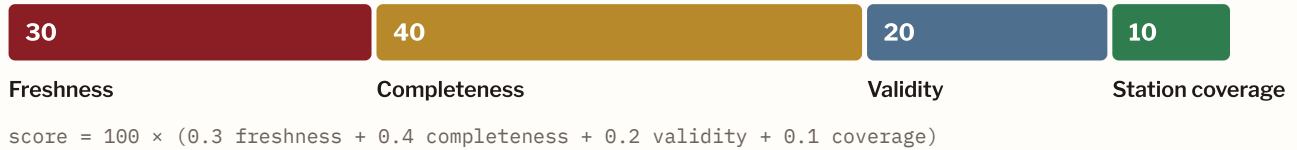
- **Severity comes from the row color.** Wepa's own green, yellow or red verdict is used directly. Codes add a name and a fix category; only a row with no color falls back to the code table.
- **Codes carry details.** `Alert_Paper_Low_Letter_530_sheets_left` is folded into `paper_low` so counts don't split by sheet count.
- **Only some codes have been seen live.** Three red codes and one yellow code are confirmed on BSU's page; the rest come from the page's legend and are marked "inferred" until they appear.
- **Levels jitter.** Sensors wobble a point or two (51, 50, 51, 50); a replacement jumps by 15 or more points.
- **The page can change.** Columns are matched by header text. A rename or reorder fails the read loudly and is logged, instead of writing values into the wrong fields.
- **Reads fail sometimes.** Network blips, Wepa maintenance and restarts all happen. Every attempt is logged, so a gap is a measured fact.

## 2.4 Status codes

| CODE                                                      | NAME                          | FALLBACK SEVERITY | FIX CATEGORY                    | SEEN ON BSU'S PA |
|-----------------------------------------------------------|-------------------------------|-------------------|---------------------------------|------------------|
| paper_out_error                                           | Paper out                     | red               | Refill / reseal paper           | Yes              |
| printer_down                                              | Printer down                  | red               | Inspect printer                 | Yes              |
| tray_missing                                              | Tray missing                  | red               | Refill / reseal paper           | Yes              |
| paper_low                                                 | Paper low                     | yellow            | Refill / reseal paper           | Yes              |
| paper_jam                                                 | Paper jam                     | red               | Clear jam                       | Inferred         |
| not_reachable                                             | Not reachable                 | red               | Check network / restart station | Inferred         |
| fatal_error ,<br>service_call                             | Fatal error,<br>service call  | red               | Power-cycle printer             | Inferred         |
| cover_open                                                | Top cover open                | red               | Inspect printer                 | Inferred         |
| toner_critical ,<br>drum_critical ,<br>toner_sensor_error | Consumable critical           | red               | Replace consumable              | Inferred         |
| toner_low ,<br>incorrect_tray_size                        | Toner low,<br>wrong tray size | yellow            | Replace consumable / paper      | Inferred         |

## 2.5 Data quality

Data quality is measured, scored and shown on the System page, so every figure can be read with its reliability in mind.



**Figure 2.1 · How the data quality score is built.** The weights are fixed in `metrics.data_quality`: freshness (time since the last good reading), completeness (readings received versus one per station per minute), validity (levels within 0 to 100) and station coverage.

**FOR DATA SCIENTISTS**

Observed time is explicit. Each reading covers the time until the station's next reading, up to 5 minutes; anything longer is unobserved and excluded from every rate. Availability is therefore a ratio of observed minutes, never of wall-clock minutes.

**FOR SUPERVISORS**

A low data-quality day means that day's numbers rest on less evidence, not that they are wrong. The System page shows which days, and the live log shows why.

# Data preparation

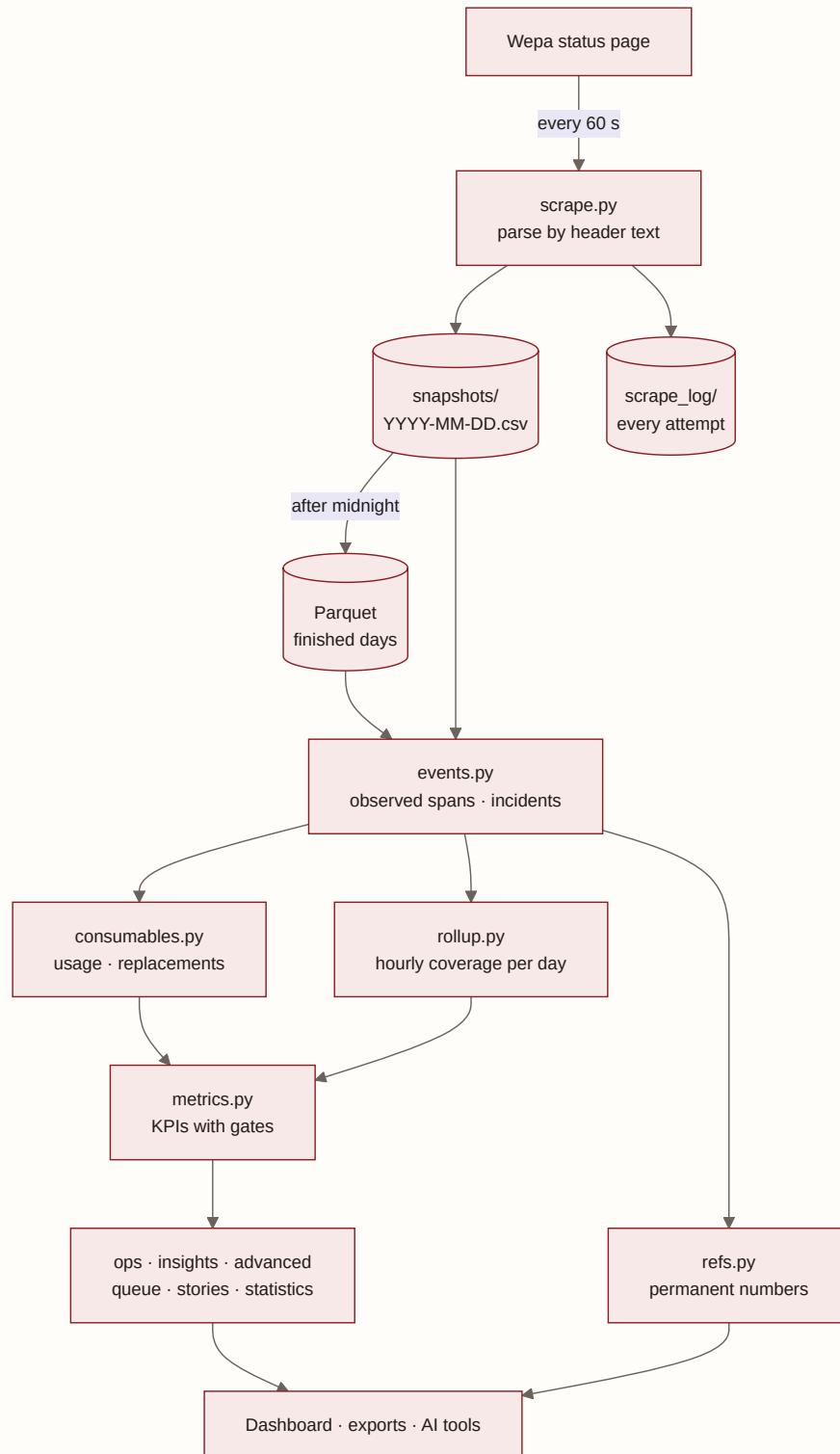
Engineers

Data scientists

## IN SHORT

Raw readings are stored exactly as read and never edited. Everything else is computed from them: when each outage started and ended, how much toner was really used, how long printers were available. If a rule changes, the whole history is simply recomputed.

## 3.1 Pipeline



**Figure 3.1 · From page to dashboard.** Storage is append-only CSV for today and Parquet for finished days. Imported history (from another computer or a backup collector) sits beside it and is merged on load, with duplicate minutes dropped.

## 3.2 Cleaning and construction rules

### One reading per minute

When two collectors overlap (a deploy, a backup), duplicates for the same station and minute are dropped; for the scrape log, a success wins over a failure.

### Observed time

A reading covers the time to the next one, up to 5 minutes. Longer gaps are unobserved: never counted as up or down.

### Incidents

Consecutive readings of one station in the same state. An incident opens at the first reading showing it and resolves at the first that doesn't. Gaps under 6 hours with the same state on both sides are bridged.

### Censored starts

An incident already open when monitoring began has an unknown start. It is kept (it is real) but excluded from repair-time averages.

### Replacements

A rise of 15 points or more is a part replacement, logged with what the old part had left, and never counted as negative usage.

### Usage

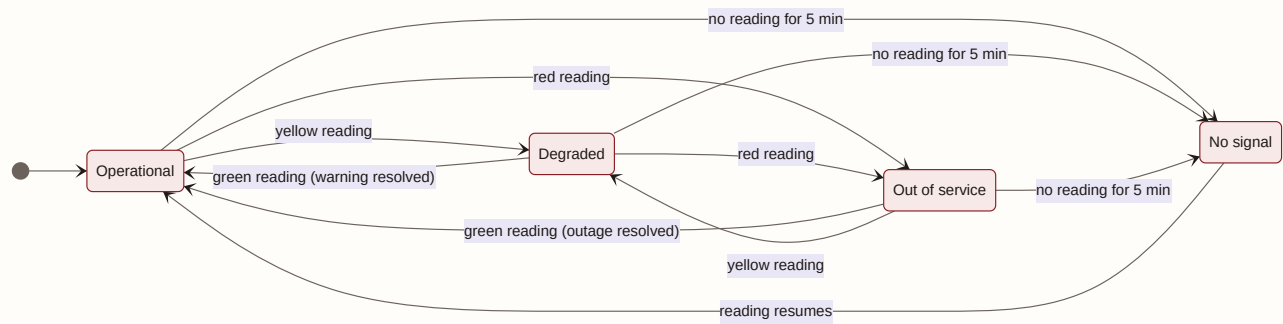
Between replacements, usage is the first reading minus the lowest reading so far, so jitter is counted once. 100 points equals one part's worth.

### Desk hours

Each outage is split into time while its team's desk was staffed and time after hours, using the configured hours and the academic calendar's holidays.

### Reference numbers

Every outage gets a permanent number by its cause: **OUT** unreachable, **JAM** jammed, **PAP** paper, **SUP** consumable out, **ERR** hardware or system; investigations **INV**, update packages **UPD**, feature requests **REQ**. Nine digits; never reused.



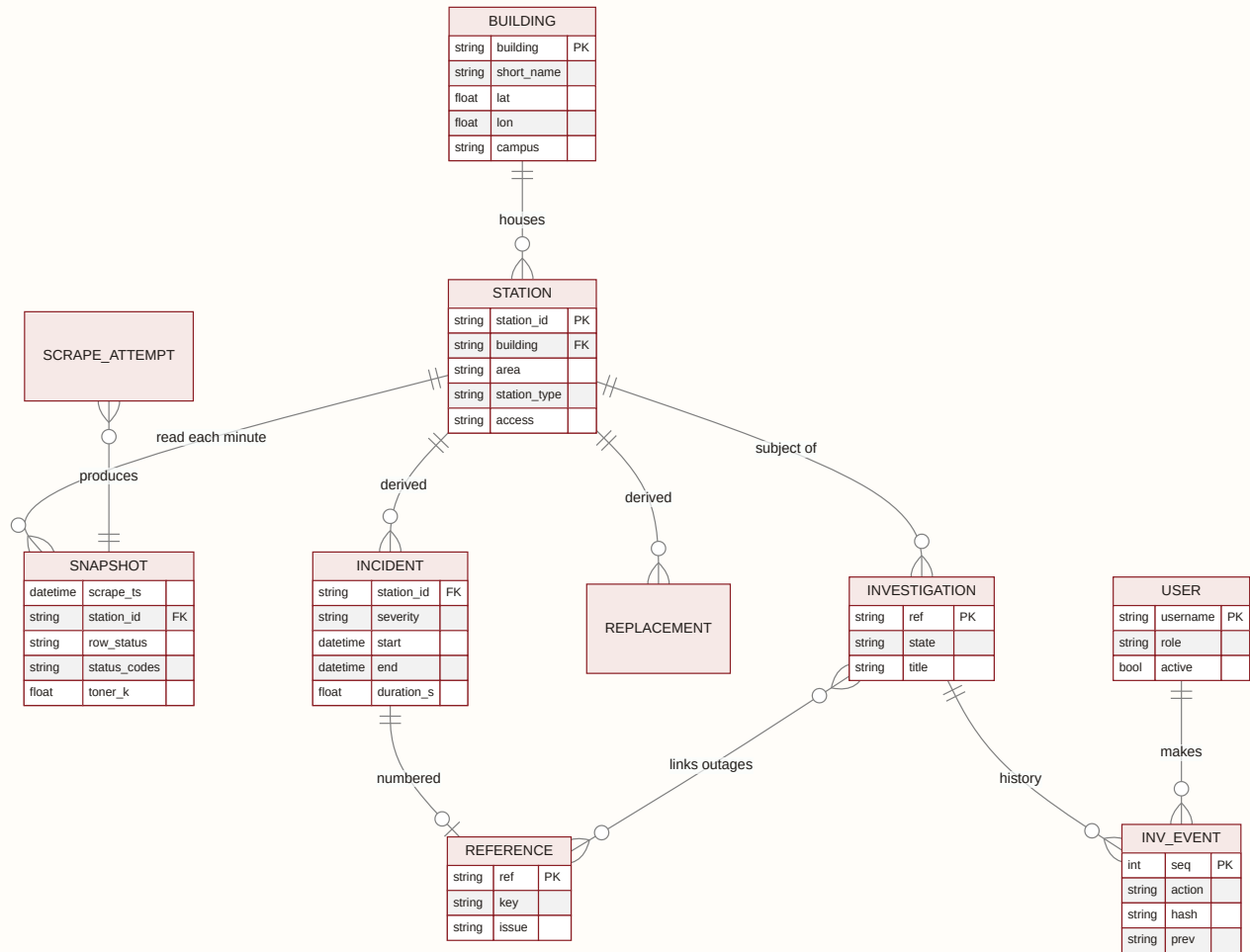
**Figure 3.2 · Station states.** The four words used everywhere. "No signal" is not a fault; it means the station's status is unknown, and that time is left out of every rate.

### 3.3 Worked example: the replacement rule

| TIME  | BLACK TONER | COUNTED AS                                        | RUNNING TOTAL |
|-------|-------------|---------------------------------------------------|---------------|
| 09:00 | 12%         | –                                                 | 0             |
| 11:00 | 7%          | usage +5                                          | 5             |
| 13:00 | 3%          | usage +4                                          | 9             |
| 13:20 | 100%        | <b>replacement</b> (3% left in the old cartridge) | 9             |
| 16:00 | 94%         | usage +6                                          | <b>15</b>     |

### 3.4 Data dictionary

Every stored field the product reads or writes. Times are UTC in storage and shown in Eastern time. Records under `records/` are append-only JSON lines unless noted.



**Figure 3.3 · Entities and relationships.** Stored entities (snapshots, attempts, references, investigation events, users) and derived ones (incidents, replacements) computed on load.

**snapshots/ · one row per station per successful read**

| FIELD         | TYPE                 | MEANING                                                         |
|---------------|----------------------|-----------------------------------------------------------------|
| scrape_ts     | datetime, UTC        | When the page was read                                          |
| page_ts       | datetime, UTC        | The timestamp printed on Wepa's page                            |
| section       | text                 | Page section: ResNet, Student Computer Labs, Satellite Campuses |
| station_id    | text, 5 digits       | Wepa's station number (P.S.#); key to the reference file        |
| description   | text                 | Wepa's station description                                      |
| row_status    | green / yellow / red | Wepa's verdict for the row                                      |
| status_codes  | text, ; - separated  | Normalized alert codes                                          |
| printer_text  | text                 | Printer message, e.g. a tray warning                            |
| toner_k c m y | number 0–100         | Toner remaining, black, cyan, magenta, yellow                   |
| drum_k c m y  | number 0–100         | Drum life remaining                                             |
| belt , fuser  | number 0–100         | Transfer belt and fuser life remaining                          |

**scrape\_log/ · one row per attempt, successful or not**

| FIELD       | TYPE          | MEANING                                    |
|-------------|---------------|--------------------------------------------|
| attempt_ts  | datetime, UTC | When the read was attempted                |
| ok          | boolean       | Whether the page was read and parsed       |
| http_status | integer       | HTTP status code, if any                   |
| duration_ms | integer       | How long the read took                     |
| n_stations  | integer       | Stations found on the page                 |
| error       | text          | Why it failed (never contains credentials) |

**reference/stations.csv · maintained by hand**

| FIELD        | TYPE                           | MEANING                                                                                                                                                                          |
|--------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| station_id   | text                           | Wepa station number                                                                                                                                                              |
| building     | text                           | Building name; key to buildings.csv                                                                                                                                              |
| area         | text                           | Campus area, e.g. University Park                                                                                                                                                |
| station_type | residence /<br>lab / satellite | What kind of place it serves                                                                                                                                                     |
| access       | public /<br>residents          | Who can walk up to it. Blank means residents for residence-hall printers and public otherwise. East Campus Commons is public : a ResNet printer in a building every student uses |
| notes        | text                           | Location detail                                                                                                                                                                  |

**Incidents (derived on load) · outages and warnings**

| FIELD                           | TYPE               | MEANING                                                        |
|---------------------------------|--------------------|----------------------------------------------------------------|
| station_id                      | text               | Station                                                        |
| severity                        | red / yellow       | Outage (out of service) or warning (degraded)                  |
| start , end                     | datetime           | First reading showing it; first reading after it cleared       |
| status                          | open /<br>resolved | Shown as Ongoing or Resolved                                   |
| censored_start                  | boolean            | Already open when monitoring began                             |
| duration_s                      | seconds            | Length, shown in days, hours and minutes                       |
| staffed_s , after_s ,<br>wait_s | seconds            | Time inside and outside desk hours; wait until the desk opened |
| ref                             | text               | Permanent reference number (outages only)                      |

## records/ · the product's own records

| FILE                                        | ONE LINE PER                    | KEY FIELDS                                                                                                                                                                                      |
|---------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>refs.jsonl</code>                     | Reference number issued         | <code>ref</code> , <code>key</code> (station and start), <code>issue</code> , <code>created</code>                                                                                              |
| <code>investigations.jsonl</code>           | Change to an investigation      | <code>seq</code> , <code>ts</code> , <code>inv</code> , <code>user</code> , <code>action</code> , <code>changes</code> (old and new), <code>note</code> , <code>prev</code> , <code>hash</code> |
| <code>system_events.jsonl</code>            | App event for the activity feed | <code>kind</code> , <code>title</code> , <code>detail</code> , <code>href</code>                                                                                                                |
| <code>versions.jsonl</code>                 | Version first run               | <code>version</code> , <code>previous</code> , <code>commit</code>                                                                                                                              |
| <code>updates.jsonl</code>                  | Update package change           | <code>ref</code> (UPD), <code>state</code> , <code>spec</code>                                                                                                                                  |
| <code>feedback.jsonl</code>                 | Feature request change          | <code>ref</code> (REQ), <code>kind</code> , <code>state</code> , <code>issue</code>                                                                                                             |
| <code>vulns.json</code> (snapshot)          | Latest vulnerability check      | findings with ids, severity, fix, priority, sources                                                                                                                                             |
| <code>security/users.json</code> (snapshot) | Staff directory                 | username, name, email, role, active, resident, hall, phone, script hash, session epoch                                                                                                          |

# Modeling

Data scientists

Supervisors

IT leadership

## IN SHORT

Most of the product's value comes from careful definitions, not exotic models: what counts as available, how long an outage really lasted, what usage really was. On top of those sit a handful of well-understood statistical methods, each chosen to answer one operational question, and one machine-learning model that has to prove itself before it is shown.

## 4.1 Measures

| MEASURE                    | DEFINITION                                                                 | SHOWN WHEN                        |
|----------------------------|----------------------------------------------------------------------------|-----------------------------------|
| Availability               | Observed printer-minutes not out of service ÷ all observed printer-minutes | At least half the window observed |
| Building coverage          | Share of minutes at least one printer in the building could print          | Same                              |
| Mean time to repair        | Mean duration of resolved outages (or warnings), censored starts excluded  | At least 3 incidents              |
| Mean time between failures | Hours of uptime ÷ number of outages                                        | At least one outage               |
| Paper refill time          | Mean duration of paper-out outages                                         | At least 3 incidents              |
| Tray empty time            | Mean time a single tray reports empty while others may still print         | At least 3 incidents              |
| Burn rate                  | Consumable points used per day, from the last 14 days                      | At least 3 days observed          |
| Data quality               | 30% freshness + 40% completeness + 20% validity + 10% station coverage     | Always                            |

Below the evidence threshold the dashboard says "insufficient data" rather than showing a number that would swing wildly with one more incident.

## 4.2 Model inventory

| QUESTION                                              | METHOD                                                                                                                                            | WHY THIS METHOD                                                        |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| When will this part reach end of life?                | Theil–Sen regression on readings since the last replacement, with a 90% interval                                                                  | Robust to sensor blips; falls back to burn rate with few readings      |
| Is today's fault count normal?                        | Statistical process control c-chart (mean $\pm 3\sqrt{\text{mean}}$ ) and the Western Electric 8-in-a-row rule; Poisson tail test per station-day | Counts of rare events; the standard Six Sigma control tool             |
| How fast are outages fixed, in and out of desk hours? | Kaplan–Meier survival curves with a log-rank test                                                                                                 | Keeps still-open outages as censored rather than dropping them         |
| Do busier printers fail more?                         | Ordinary least squares, outages per week on usage, with residual flags beyond $2\sigma$                                                           | Separates "busy" from "bad unit"                                       |
| Which printers fail most, fairly?                     | Empirical-Bayes Gamma–Poisson rates with 90% credible intervals; report-card grades shrink toward the campus rate                                 | Small samples don't top the list by chance                             |
| Did several printers fail together?                   | Clusters of outages within 10 minutes across 3+ stations, tested against a Poisson rate for that hour and day type                                | Tells a network or Wepa-side event from a busy moment                  |
| Which downtime cost the most printing?                | Outage hours weighted by that printer's typical use at that time of week                                                                          | An hour down at 2 PM on a Tuesday matters more than at 4 AM            |
| Does humid weather go with more jams?                 | Mantel–Haenszel rate ratio, stratified by 6-hour block and weekday or weekend                                                                     | Controls for the fact that humidity and printing both follow the clock |
| What if a desk stayed open later?                     | Monte Carlo replay (1,000 runs) of real outages with extra desk coverage; fix times drawn from real desk-hour fixes                               | Answers staffing questions with the campus's own history               |
| Where would one more printer help most?               | Greedy one-step p-median on walking times, weighted by busy-weighted downtime                                                                     | Directly minimizes students' extra walking                             |
| Supplies to reorder                                   | Monte Carlo of recent daily usage: how many of each part to stock at 50%, 90% and 95%                                                             | Gives a range, not a single guess                                      |

confidence; reorder points use the supply lead time (7 days unless set)

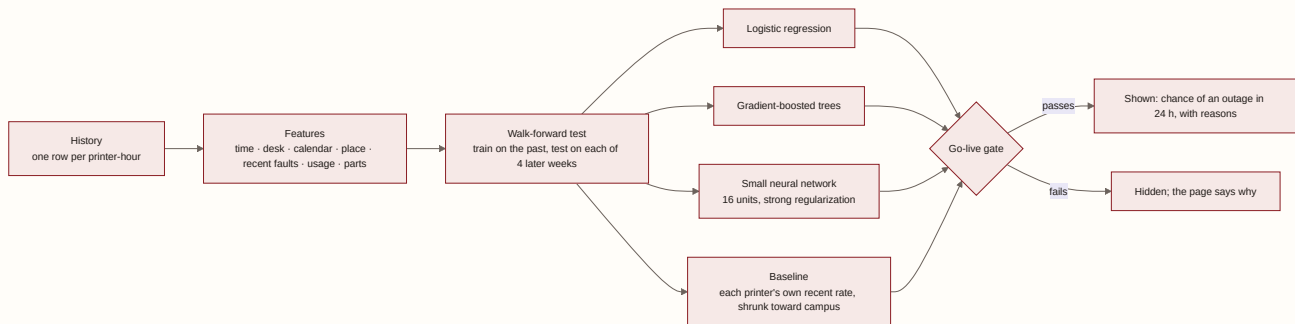
Which printer may fail in the next 24 hours?

Machine learning, below

Only if it beats the simple alternative

## 4.3 The outage-risk model

Each printer-hour becomes one example: time of day and week, whether a desk is open and how long until it opens, whether classes are in session, the kind of place, recent faults and warnings, recent usage, part levels. The label is whether an outage started in the next 24 hours.



**Figure 4.1 · Train, test, gate.** Recent history counts more (weight halves every 21 days). The model retraines nightly; every prediction is logged so its track record can be checked later.

## 4.4 Decision tables (DMN)

The product's key decisions are written as DMN-style decision tables. The hit policy is shown in the top-left cell: **U** unique (exactly one rule applies), **F** first (the first matching rule wins), **A** all conditions must hold.

**Decision: station status**

| <b>F</b> | <b>LAST READING AGE</b> | <b>ROW COLOR</b> | <b>STATUS SHOWN</b> |
|----------|-------------------------|------------------|---------------------|
| 1        | over 5 minutes          | –                | No signal           |
| 2        | ≤ 5 minutes             | red              | Out of service      |
| 3        | ≤ 5 minutes             | yellow           | Degraded            |
| 4        | ≤ 5 minutes             | green            | Operational         |

**Decision: show the outage-risk model**

| <b>A</b> | <b>CONDITION</b>                      | <b>THRESHOLD</b> |
|----------|---------------------------------------|------------------|
| 1        | Days of history                       | ≥ 21             |
| 2        | Outages in training                   | ≥ 50             |
| 3        | Outages in the test weeks             | ≥ 25             |
| 4        | Ranking skill (AUC) of the best model | ≥ 0.65           |
| 5        | Brier skill over the baseline         | ≥ 2%             |

**Decision: vulnerability priority**

| <b>F</b> | <b>EVIDENCE FROM THE SOURCES</b>                                                                               | <b>PRIORITY</b>                       |
|----------|----------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 1        | On CISA's Known Exploited list, or CISA or Microsoft report active exploitation, or a Metasploit module exists | Act now (Soon for install-only tools) |
| 2        | Public exploit on Exploit-DB, CISA proof of concept, automatable, EPSS ≥ 10%, or high or critical severity     | Soon                                  |
| 3        | None of the above                                                                                              | Routine                               |

**Decision: investigation impact**

| <b>F</b> | <b>HISTORY</b>  | <b>SCORE (USAGE + REDUNDANCY + LOST PRINTING, EACH 1–3)</b> | <b>IMPACT</b>      |
|----------|-----------------|-------------------------------------------------------------|--------------------|
| 1        | under 90 days   | –                                                           | Moderate (default) |
| 2        | 90 days or more | average under 1.5                                           | Low                |
| 3        | 90 days or more | average 2.5 or more                                         | High               |
| 4        | 90 days or more | otherwise                                                   | Moderate           |

**Decision: backup collector**

| <b>U</b> | <b>BACKUP STATE</b> | <b>MAIN COLLECTOR</b> | <b>ACTION</b>                                                      |
|----------|---------------------|-----------------------|--------------------------------------------------------------------|
| 1        | Standing by         | live                  | Stay on standby; update itself if the main app is newer            |
| 2        | Standing by         | silent under 60 s     | Keep checking every 15 s                                           |
| 3        | Standing by         | silent 60 s or more   | Take over: collect every minute, send readings to the dashboard    |
| 4        | Collecting          | live again            | Stop at once, report the period covered, send everything collected |
| 5        | Collecting          | still silent          | Keep collecting                                                    |

## 4.5 The AI analyst

The Assistant is optional and runs only with an API key. It is an analyst with 16 read-only tools over the same data the pages use (station profiles, availability, incidents, repair times, supplies, usage, the report card, period comparisons, campus context, statistics, the risk model, investigations and the app's self-check).

- **Facts only.** Every number it writes is checked against the tool results it received, allowing only honest unit restatements (minutes to hours, shares to percentages). Unsupported figures trigger one rewrite; anything still unsupported is flagged.

- **Read-only.** It cannot change records, start updates or contact anyone.
- **Safe output.** Answers are shown without links, images or HTML.
- **Scoped and fair.** BSU printing only; never guesses who caused a problem; says "demo data" when it is.

# Evaluation

IT leadership

Data scientists

Engineers

## IN SHORT

Each output is checked three ways: does the code do what the definition says (automated tests), is the evidence strong enough to show it (gates), and does the model actually beat the obvious alternative (held-out testing). Where the answer is no, the product says so instead of showing a confident number.

## 5.1 Does it meet the business goals?

| OBJECTIVE                           | EVIDENCE IN THE PRODUCT                                        | ASSESSMENT                                             |
|-------------------------------------|----------------------------------------------------------------|--------------------------------------------------------|
| Fix printers sooner                 | Outage detected at the first red reading; ranked queue; routes | <b>Met in design</b> Outcome needs a term of live data |
| Fewer outages waiting overnight     | Desk-hours split per team; staffing what-if                    | <b>Met</b>                                             |
| Evidence for the vendor             | Reference numbers; detector; hash chain; evidence PDF          | <b>Met</b>                                             |
| Supplies before they run out        | Replacement-aware usage; end-of-life and reorder forecasts     | <b>Met</b> Accuracy improves with history              |
| Point students to a working printer | Walking network; access rules; QR status pages                 | <b>Met</b>                                             |
| Trustworthy numbers                 | Quality score; gates; no imputation; definitions               | <b>Met</b>                                             |

## 5.2 How the outage-risk model is scored

### Walk-forward

Train on everything before a week, test on that week, for each of the last four weeks. The model never sees the future it is tested on.

|                     |                                                                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Brier score</b>  | Mean squared error of the predicted probabilities; lower is better.<br><b>Skill</b> is the improvement over the baseline's Brier score. |
| <b>AUC</b>          | How well the model ranks printers that did fail above those that didn't; 0.5 is a coin flip.                                            |
| <b>Calibration</b>  | Predicted chance versus observed rate in bins, shown as a chart, so "20%" means about 1 in 5.                                           |
| <b>Track record</b> | Live predictions are logged and later compared with what happened.                                                                      |

#### FOR DATA SCIENTISTS

The baseline is deliberately strong: each printer's recent outage rate, shrunk toward the campus rate. An early version's baseline scored below 0.5 AUC because raw per-printer rates were noisy; shrinkage fixed it. A model that cannot beat this baseline is not shown.

#### FOR LEADERSHIP

"Not enough evidence yet" is a correct answer. The risk model stays hidden on new deployments until the data supports it; the page explains which condition is not met.

## 5.3 Software verification

- **190 automated test cases** run on every push and daily: the page parser against saved real pages, incident rules, the replacement rule, desk hours, the calendar, stories, Ask the data, routing (including an exact Held-Karp check), investigations, accounts and sign-in, security probes, the update pipeline, backup collectors, threat intelligence and durations.
- **Lint** (ruff), **static security analysis** (bandit) and **known-vulnerability scanning** (pip-audit over the exact lock file) in continuous integration.
- **Attacker-style tests**: every route without a session, path traversal, oversized uploads, cross-site posts, password guessing, forged or replayed backup messages, tampering with the audit trail.

## 5.4 Known limits

- Usage is inferred from consumable levels; Wepa's job counts are not available.

- Several alert codes are inferred from Wepa's legend until they appear on BSU's page.
- Forecasting across semesters (ARIMA and similar) is deliberately postponed until a full academic year of live history exists.
- Demo data is synthetic. On it, the models mostly rediscover patterns built into the simulator.
- No multi-factor authentication yet; the administrator is a shared account.

# Deployment

Engineers

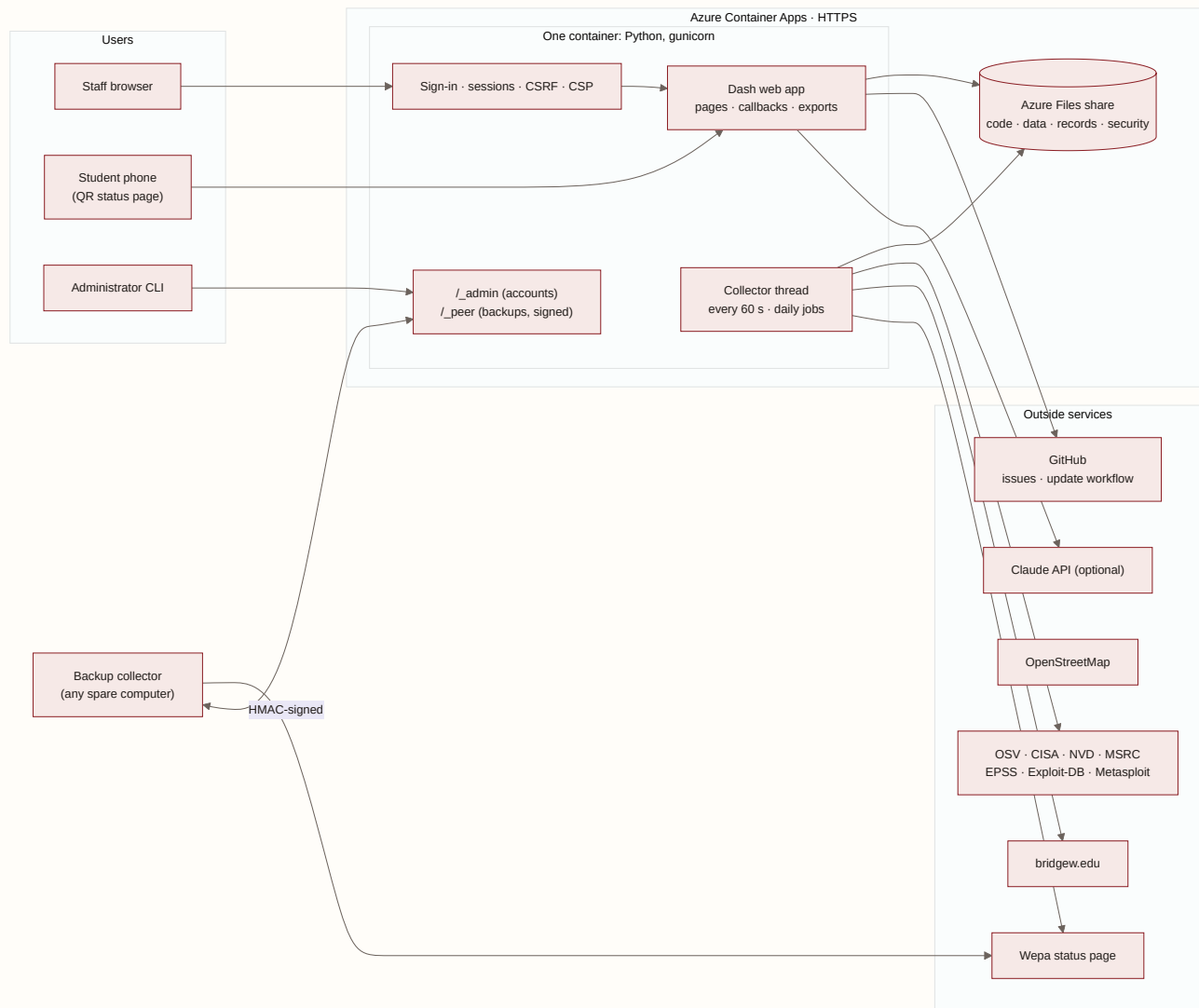
IT leadership

## IN SHORT

The app runs in Microsoft Azure as one container that both collects and serves the website, with its data on a file share that survives restarts.

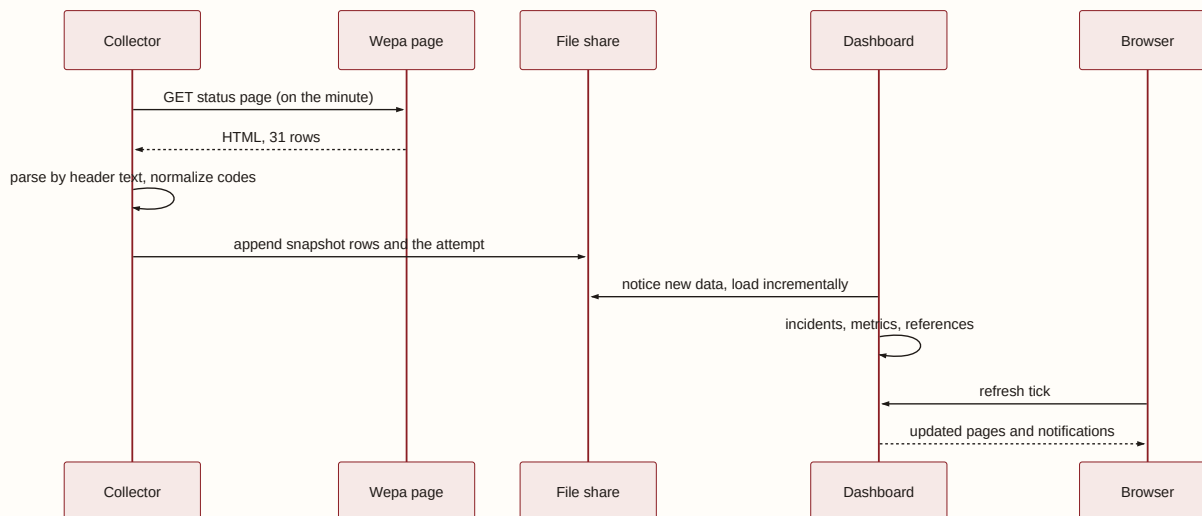
Updates roll out without taking the site down. A second computer can stand by to collect if the main collector ever stops, and the app checks its own health and security every day.

## 6.1 Architecture



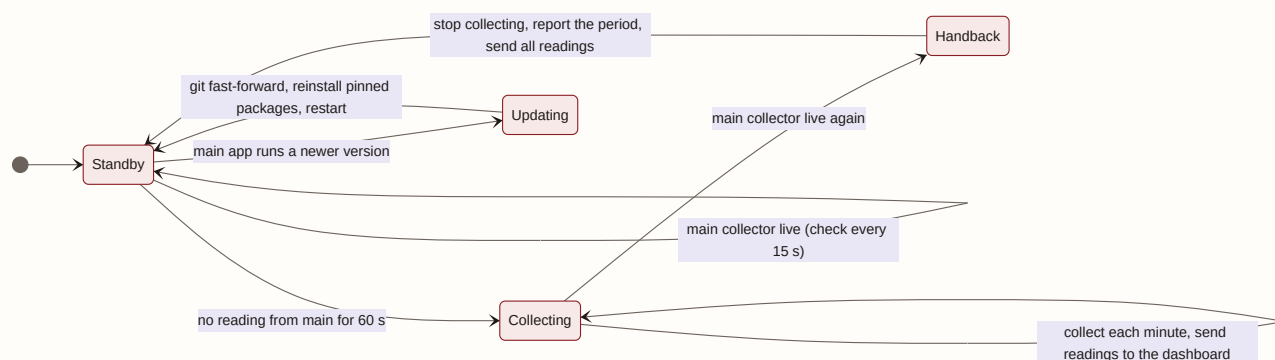
**Figure 6.1 · Containers and connections.** One container runs both the website and the collector; a lock on the share ensures only one copy collects. OpenStreetMap data is fetched once and kept.

## 6.2 Every minute



**Figure 6.2 • The one-minute cycle.** Once a day the collector also compacts finished days, refreshes campus data and weather, retrain the risk model, checks for vulnerabilities and sends queued feature requests.

## 6.3 Staying up



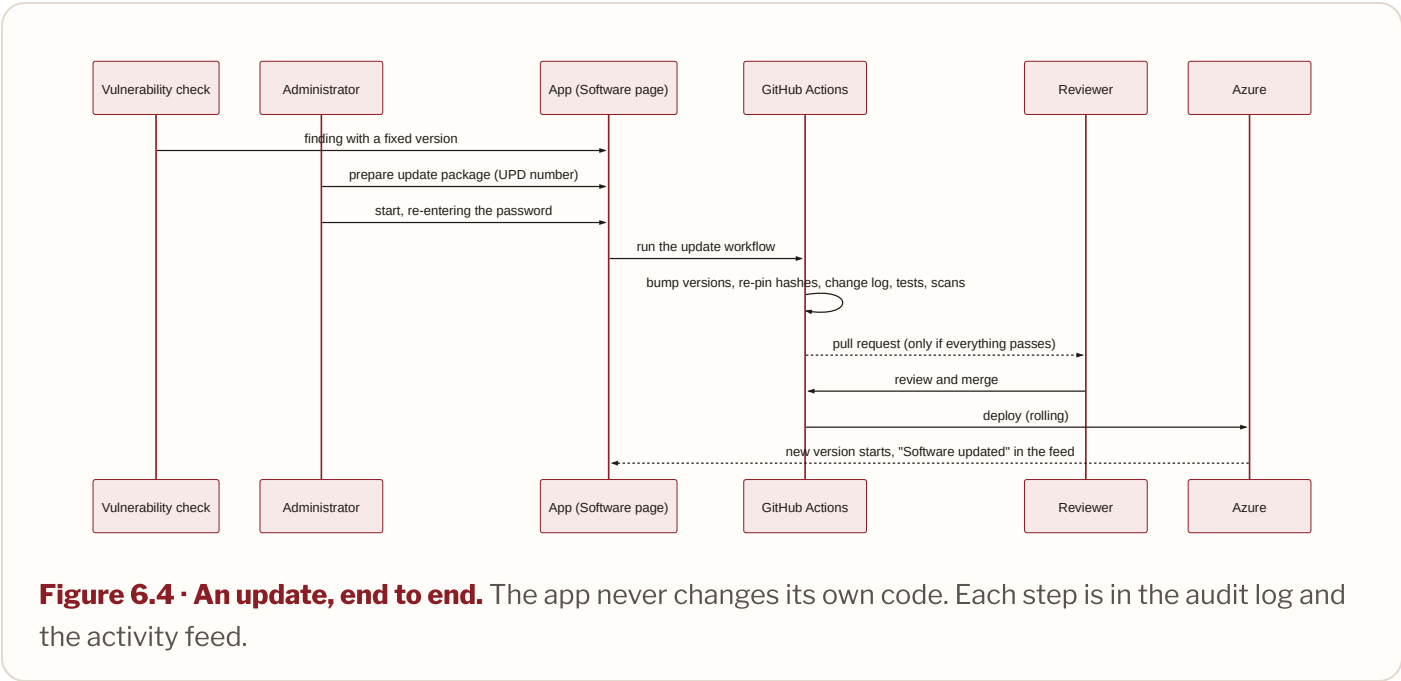
**Figure 6.3 • Backup collector.** Every message is signed with HMAC-SHA256 over method, path, time, a one-time number and the body; replies are signed too, so an impostor cannot make a backup stand down.

# 6.4 Security: confidentiality, integrity, availability

| PRINCIPLE       | CONTROLS                                                                                                                                                                                           |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Confidentiality | Sign-in on every page; named accounts with roles; sessions end after 2 hours idle or 12 hours; NIST SP 800-63B passwords with breached-password refusal; lockouts; secrets stored as Azure secrets |
| Integrity       | Append-only readings; hash-chained investigation history; CSRF tokens; Content Security Policy; hash-pinned package installs; signed backup messages                                               |
| Availability    | Rolling deploys; collector lock; backup collectors; optional second replica; gaps recorded, never guessed                                                                                          |

The Software page carries a live OWASP Top 10 (2025) checklist with evidence for each control, and `docs/SECURITY.md` holds the threat model, attacker-style test results and known limits.

# 6.5 Updates



## 6.6 Monitoring and maintenance plan

| WHAT                                  | HOW OFTEN               | WHERE TO LOOK                              | ACTION WHEN WRONG                                    |
|---------------------------------------|-------------------------|--------------------------------------------|------------------------------------------------------|
| Collector freshness, data quality     | Continuous              | System page, self-check                    | Read the live log; restart the app; check the backup |
| Known vulnerabilities                 | Daily                   | Software page                              | Prepare an update package for "Act now" first        |
| Investigation audit trail             | Every self-check        | Software page                              | Restore records from the share's backup              |
| Risk model gate and track record      | Nightly retrain         | Analytics → Forecasts and statistics       | None needed; the gate hides a weak model             |
| Reference files (stations, buildings) | When the campus changes | Stations page shows "Unassigned"           | Edit <code>reference/*.csv</code> , deploy           |
| Desk hours, holidays                  | Each term               | <code>config.py</code> ; academic calendar | Update and deploy                                    |

# How the work flows

Supervisors

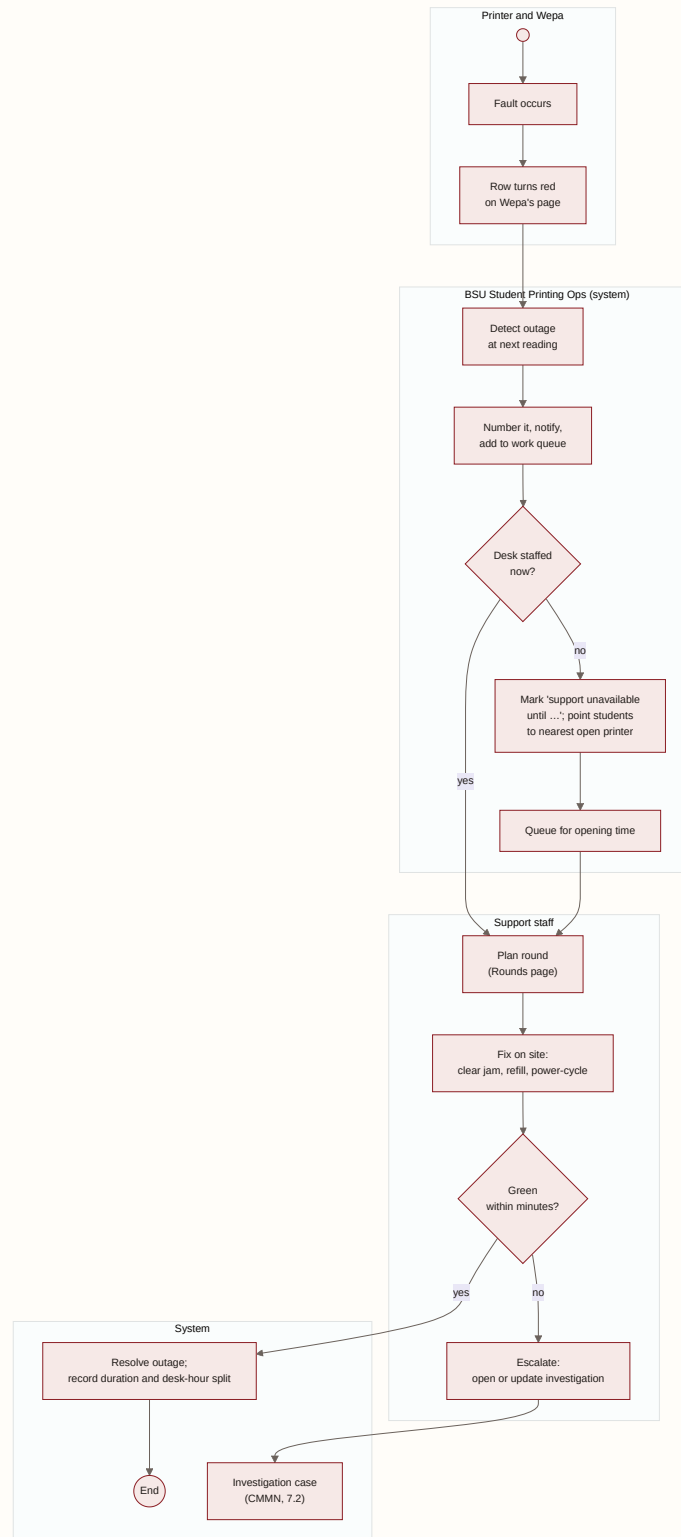
IT leadership

Student staff

## IN SHORT

Three standard notations describe the work around the product: BPMN for the routine outage process, CMMN for investigations (which don't follow a fixed path), and DMN for decisions (chapter 4.4). A Six Sigma view shows how the product supports continuous improvement.

## 7.1 Outage to resolution (BPMN-style)

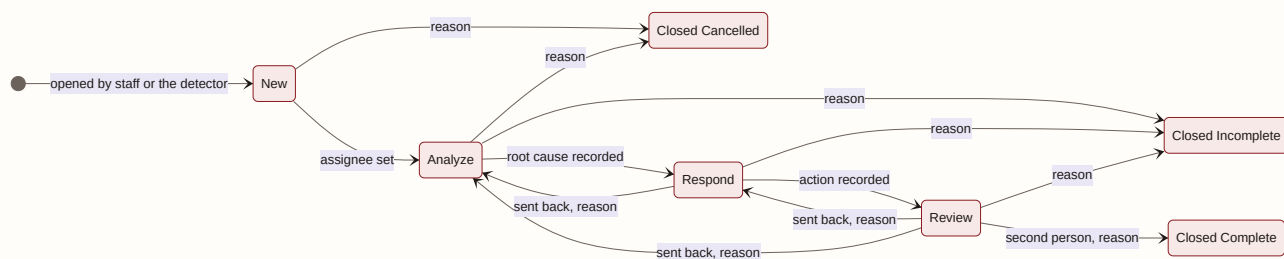


**Figure 7.1 · Outage process in four lanes.** Circles are start and end events, rounded boxes tasks, diamonds gateways, as in BPMN 2.0. Drawn with flowchart tooling, so shapes approximate the standard.

## 7.2 Investigations as a case (CMMN view)

An investigation is knowledge work: people decide what to do next. CMMN models it as a case file with stages, entry criteria (sentries) and milestones, rather than a fixed sequence.

| CMMN ELEMENT        | IN THE PRODUCT                                                                                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Case file           | The investigation record: title, station, category, impact, assignee, root cause, actions, ITSM and Wepa case numbers, linked outages, notes |
| Stage: Analyze      | Entry sentry: an assignee is set (also sets the escalation date)                                                                             |
| Stage: Respond      | Entry sentry: a root cause, marked suspected or confirmed                                                                                    |
| Stage: Review       | Entry sentry: the action taken is recorded. Exit needs a different person (separation of duties)                                             |
| Milestones          | Closed Complete; Closed Incomplete; Closed Cancelled (each needs a written reason)                                                           |
| Discretionary tasks | Add a note; link outages; download the evidence packet; override impact                                                                      |
| Event listener      | The detector opens a case when a printer has 3 or more unreachable or hardware outages in 14 days                                            |
| Case file integrity | Every change is an event with the previous event's SHA-256 hash; archived 2 years after closing                                              |



**Figure 7.2 · Investigation lifecycle.** Fields are editable only in New, Analyze and Respond. Time in each state, send-backs and reassignments are measured on the Investigations page.

## 7.3 Six Sigma: DMAIC with the product

| PHASE   | QUESTION                                   | WHERE THE PRODUCT ANSWERS IT                                                                                         |
|---------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Define  | What does a defect look like to a student? | An outage: a station that cannot print. CTQs in figure 1.2                                                           |
| Measure | How often, how long, where, when?          | Availability, repair time, time between failures, desk-hour split, data quality                                      |
| Analyze | Why?                                       | Fault ranking by cause and hour, shared outages, busy-weighted downtime, humidity test, Kaplan–Meier, investigations |
| Improve | What would help most?                      | Staffing what-ifs, printer placement, reorder points, vendor escalation with evidence                                |
| Control | Did it stay better?                        | c-chart with run rules, period comparisons with change tests, Executive summary                                      |

# Using BSU Student Printing Ops

[Students](#)[Student staff](#)[Supervisors](#)[IT leadership](#)

## IN SHORT

Sign in with the account your ResNet administrator gave you. The menu groups the pages by what you came to do (Operate, Analyze, Report, Manage). The header holds Alerts, the Assistant and your account menu; Filters and Download sit under each page's title. Staff record investigations, inventory and the key log under their own names; everything else is read-only.

## 8.1 For students: finding a printer that works

- 1 Scan the QR sign on the printer, where your residence hall or building has one.
- 2 The page shows whether that printer is working right now and, if not, the nearest printers that are, with the walking time. East Campus Commons counts as open to everyone; other residence-hall printers need card access.
- 3 Wepa holds print jobs online, so a job you sent can usually be released at another working station.

Student status pages are behind sign-in unless the administrator turns on public status pages (with Wepa's and BSU's agreement). They show the present only, never history.

## 8.2 Getting started (staff)

- 1 Open the app's address and sign in. A temporary password from the administrator must be changed the first time.
- 2 Use a password of 12 or more characters; a few unrelated words works well. Common and breached passwords are refused.

- 3 The header has three buttons: **Alerts** (stations going down or coming back, with a link to the full activity log), the **Assistant**, and your **account menu** (My account, Appearance, Directory, Suggest a feature, the administration pages, Sign out).
- 4 Under **Appearance**, choose BSU, light, dark or automatic, larger text, compact spacing, reduced motion or extra contrast. On a BSU football game day the BSU appearance becomes **Go Bears**.

## 8.3 Finding your way

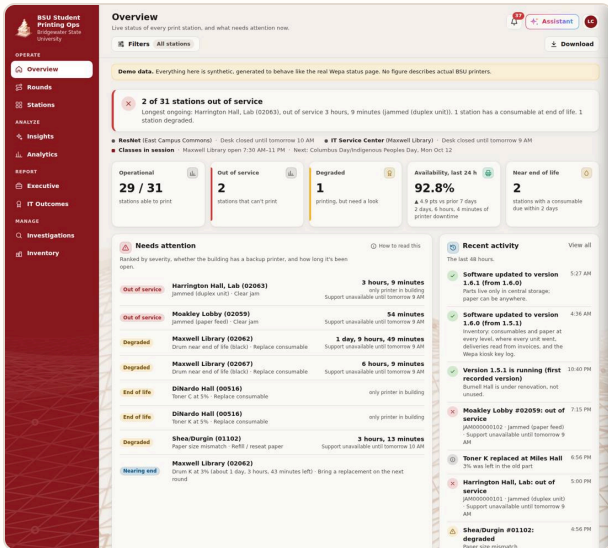
The menu is grouped by what you came to do. On a computer it runs down the left; on a phone the four most used pages sit along the bottom and **More** opens the rest.

| GROUP                         | PAGES                                                |
|-------------------------------|------------------------------------------------------|
| Operate                       | Overview · Rounds · Stations                         |
| Analyze                       | Insights · Analytics                                 |
| Report                        | Executive summary · IT Outcomes                      |
| Manage                        | Investigations · Inventory                           |
| Administration (account menu) | System health · Software and security · Activity log |

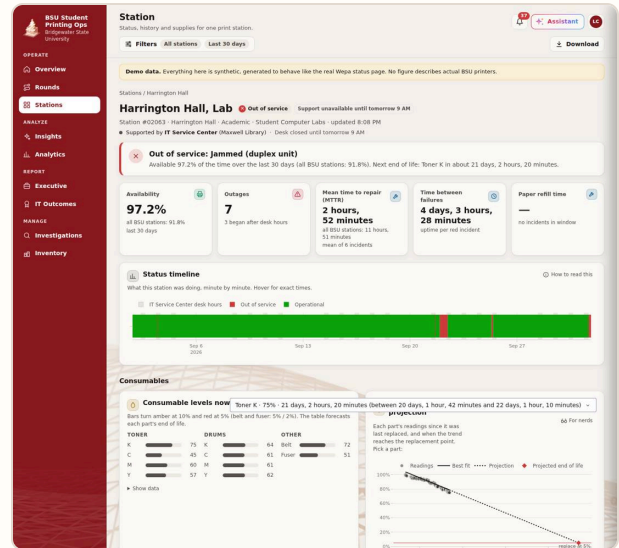
**Filters** sit under the page title, in the same place on every page that has them. The button shows what's applied (for example "All stations · Last 30 days"); open it to choose stations and the period, or **Clear filters**. **Download**, beside it, exports the data behind the page. Tabs along the top of a page (Analytics, Investigations, Inventory) go into the address, so a tab can be bookmarked or shared.

## 8.4 Pages

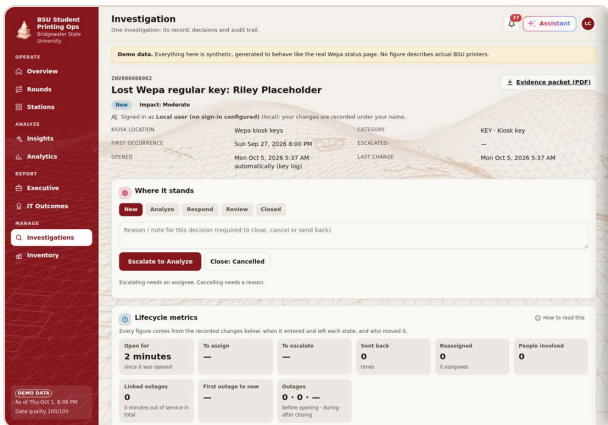
| PAGE                  | USE IT TO                                                                                                                                    | TIP                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Overview              | See what needs attention now: status map, work queue, end-of-life watch, recent activity                                                     | Start every shift here                                                                 |
| Rounds                | Get the fastest route to every printer that needs a visit, on foot or by van, from a Home Base or supply closet; choose accessible entrances | The administrator draws doors, printers, parking, the vans and paths in the map editor |
| Stations              | Search all stations; open one for its history, parts and backups                                                                             | Select any chart point for its meaning                                                 |
| Insights              | Read the story of a day, week or term; ask the data a question in plain English                                                              | Good for a supervisor's weekly note                                                    |
| Analytics             | Reliability, faults, supplies, usage, report card, planning, forecasts and statistics                                                        | Set the period under Filters                                                           |
| Executive summary     | Month and year to date, with what changed                                                                                                    | Download as PDF for meetings                                                           |
| IT Outcomes           | A print-ready feature for the IT annual report; in the launch year, a stakeholder briefing                                                   | Prints on letter paper                                                                 |
| Investigations        | Document recurring problems, missing stock and lost keys; move them through the lifecycle; download evidence for Wepa                        | A second person closes from Review                                                     |
| Inventory             | Stock and its trail; record deliveries, moves, counts and write-offs; weekly paper checks; the kiosk key log; storage setup                  | Deliveries need a second person's approval                                             |
| System health         | Is the monitor healthy? Live log, data quality, visits, sign-in security, AI status                                                          | Check here if numbers look odd                                                         |
| Software and security | Version, self-check, vulnerabilities and their priority, OWASP checklist, updates, backups, change log                                       | The administrator starts updates here                                                  |
| Activity log          | Everything that happened, searchable by date                                                                                                 | Select a version number for its change log                                             |
| Assistant             | Ask questions in plain English; answers cite only looked-up figures                                                                          | Ctrl or Cmd + I opens it                                                               |



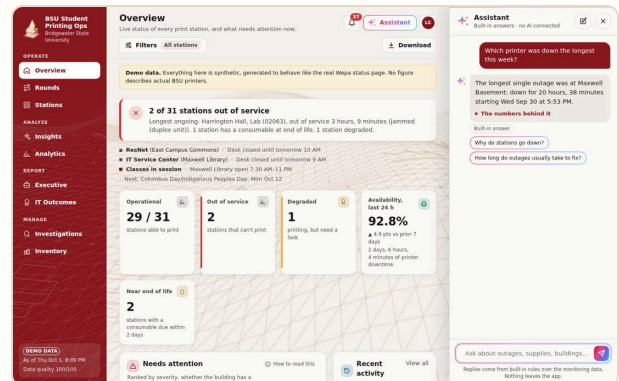
**Overview.** Status map, work queue and what needs attention.



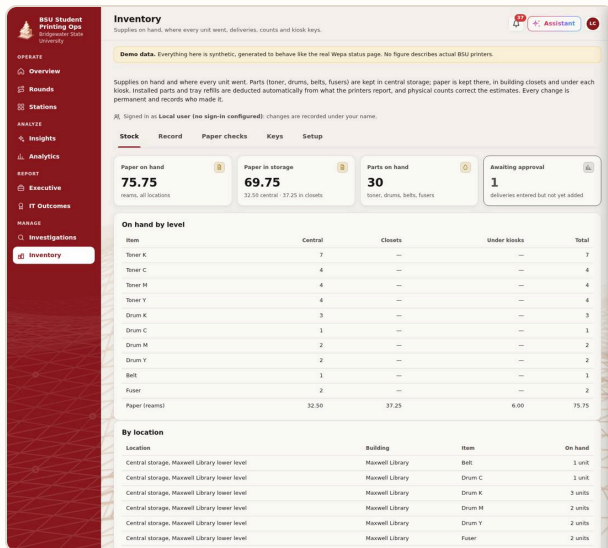
**Station page.** History, parts and the nearest backups.



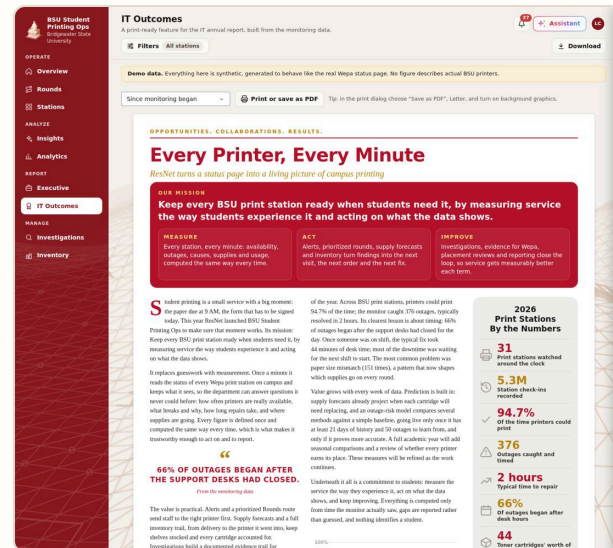
**Investigation.** Governed lifecycle, audit trail and evidence packet.



**Assistant.** Answers from read-only tools, fact-checked.



**Inventory.** Stock at every level and the trail of every unit.



**IT Outcomes.** The launch-year stakeholder briefing.

Screenshots use the synthetic demo data set.

## 8.5 Common tasks

### RESPOND TO AN OUTAGE

- 1 On Overview, open the top item in the work queue, or go to Rounds for a route.
- 2 Fix it on site. The station turns Operational within a minute of Wepa's page updating.
- 3 The outage resolves itself with its reference number, duration and desk-hour split. Nothing to type.

### WORK AN INVESTIGATION

- 1 Open Investigations. Detector-opened cases are marked; you can also open one yourself.
- 2 Assign it (moves to Analyze), record a root cause (Respond), record the action taken (Review).
- 3 A second person closes it. Download the evidence packet (PDF) for a Wepa case.

## REPORT THE MONTH

- 1 Open **Executive**, pick the month, read the highlights.
- 2 Use **Download**, beside **Filters**, for PDF, Excel, CSV or JSON of the current filters and period.

## RECEIVE A DELIVERY OF SUPPLIES

- 1 Open **Inventory** › **Record** › **Delivery received** and upload the invoice or a photo of the receipt. The lines fill in; check each against the boxes, or type them.
- 2 Submit. A second person checks it and approves it; only then is it added to stock.

## COUNT PAPER (WEEKLY, BY FRIDAY)

- 1 Open **Inventory** › **Paper checks** to see what's due, then **Record** › **Counted**.
- 2 Enter reams to two decimals (a half-used ream is 0.50). If more than half a ream, or any part, is missing, an Inventory investigation opens to find out where it went.

## A WEPA KEY IS LOST

- 1 Open **Inventory** › **Keys** › **Key returned, lost or found**, choose the key and **Lost**, and say what happened.
- 2 A Kiosk key investigation opens. It requires a counseling conversation with management, recorded on the investigation; for a master key, decide with Wepa whether the locks must change.

## SUGGEST A FEATURE OR REPORT A PROBLEM

- 1 In your account menu, choose **Suggest a feature**.
- 2 Pick the kind, give a title and details, and send. It gets a REQ number and becomes a GitHub issue; your name is not included unless the administrator turned that on.

## 8.6 For the administrator

| TASK                     | COMMAND OR PLACE                                                                                             |
|--------------------------|--------------------------------------------------------------------------------------------------------------|
| Add a staff account      | <code>python -m wepa_monitor users add jsmith --name "Jordan Smith" --email jsmith@bridg</code>              |
| Reset, deactivate, list  | <code>users reset jsmith</code> · <code>users deactivate jsmith</code> · <code>users list</code>             |
| Deploy or update         | <code>./deploy/azure-containerapps/deploy.sh</code> (from a laptop or Azure Cloud Shell)                     |
| Add keys safely          | <code>ASK_SECRETS=1 ./deploy/azure-containerapps/deploy.sh</code> (hidden prompts)                           |
| Check the app            | <code>python -m wepa_monitor selfcheck --vulns</code> , or the Software page                                 |
| Start a backup collector | <code>python -m wepa_monitor backup --primary https://&lt;app&gt;</code> with <code>WEPA_PEER_KEY</code> set |
| Start a security update  | Software page → Prepare update package → Start update (password again)                                       |

## 8.7 Frequently asked questions

**Why does a station say "No signal"?**

It hasn't appeared on Wepa's page for 5 minutes. Its status is unknown, and that time is left out of every rate.

**Why "support unavailable until ..."?**

The team that supports that printer has no desk open right now; the time is when it next opens.

**Why does a figure say "insufficient data"?**

Too few incidents or too little observed time to be reliable. It appears once there is enough evidence.

**Why isn't the outage-risk model shown?**

It hasn't beaten the simple baseline on held-out weeks yet; the page says which condition is not met.

**Is "In progress" ever shown?**

No. The monitor can see that a problem is open, not whether someone is working on it, so it says Ongoing.

# Standards and recommendations

IT leadership

Engineers

Data scientists

## IN SHORT

The product was built with recognized standards and professional codes in mind. This chapter maps each one to what the product does, says honestly where it falls short, and recommends next steps. It is a self-assessment: no certification or formal audit is claimed.

## 9.1 Alignment matrix

| STANDARD OR CODE                                  | WHAT IT ASKS, IN BRIEF                                                            | HOW THE PRODUCT ALIGNS                                                                         | STATUS  |
|---------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------|
| CRISP-DM                                          | Six phases from business understanding to deployment, iterated                    | This document's structure; nightly retraining and self-checks keep the cycle running           | Aligned |
| ISO/IEC 25010 (product quality)                   | Quality characteristics such as reliability, security, maintainability, usability | Quality requirements NFR-01 to NFR-12, each with a mechanism                                   | Aligned |
| ISO/IEC/IEEE 26514 (information for users)        | Audience analysis, task-oriented content, consistent terminology                  | Reading paths by audience; task steps; one status vocabulary; glossary                         | Aligned |
| ISO/IEC/IEEE 15289 (life-cycle information items) | Defined content for plans, descriptions and records                               | Document control, requirements, architecture, change log, test evidence                        | Partial |
| ISO 8601                                          | Unambiguous dates and times                                                       | Stored in UTC ISO 8601; shown in local time with the time zone named                           | Aligned |
| ISO/IEC 27001 (information security)              | Risk-based security controls and management                                       | Access control, cryptography, logging, secure development, supplier checks; no formal ISMS     | Partial |
| OWASP Top 10 (2025)                               | The ten most critical web application risks                                       | Live checklist with evidence in the app: 7 met, 3 partial on a correctly configured deployment | Partial |
| NIST SP 800-63B                                   | Password and session guidance                                                     | Length over complexity, breached-password check, no forced expiry, lockouts; MFA not yet       | Partial |
| BPMN 2.0, CMMN 1.1, DMN 1.x (OMG)                 | Notations for processes, cases and decisions                                      | Figures 7.1 and 7.2 and the decision tables in 4.4                                             | Aligned |
| Six Sigma (DMAIC, SIPOC, SPC)                     | Define and measure defects; control with statistics                               | SIPOC, CTQ tree, DMAIC map, c-chart with run rules                                             | Aligned |
|                                                   |                                                                                   |                                                                                                |         |

|                                                 |                                                                                    |                                                                                   |         |
|-------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------|
| ACM Code of Ethics (2018)                       | Contribute to society, avoid harm, be honest, respect privacy, keep systems secure | Mapped in 9.2                                                                     | Aligned |
| ASA Ethical Guidelines for Statistical Practice | Integrity of data and methods; responsibilities to stakeholders and data subjects  | Mapped in 9.3                                                                     | Aligned |
| DASCA role families                             | Data engineering, analysis and science as distinct competencies                    | Mapped in 9.4                                                                     | Aligned |
| Data ecology                                    | Data as an ecosystem of sources, stewards, flows and uses                          | Mapped in 9.5                                                                     | Aligned |
| WCAG 2.2                                        | Accessible web content                                                             | Keyboard focus, contrast option, reduced motion, text sizes; not formally audited | Partial |

## 9.2 ACM Code of Ethics

| PRINCIPLE                                                                          | IN PRACTICE                                                                           |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1.1 Contribute to society and human well-being                                     | The product's goal is that students can print when they need to                       |
| 1.2 Avoid harm                                                                     | Data is about printers, never people; the AI never guesses who caused a problem       |
| 1.3 Be honest and trustworthy                                                      | No imputed data; evidence gates; demo data labelled on every page                     |
| 1.6 Respect privacy                                                                | Visits shortened to the network; no passwords logged; names kept out of public issues |
| 2.5 Give comprehensive evaluations, including risks                                | Chapter 5; known limits stated in the app and here                                    |
| 2.9 Design and implement systems that are robustly and usably secure               | Chapter 6.4; docs/SECURITY.md; attacker-style tests                                   |
| 3.7 Recognize and take special care of systems that become integrated into society | Vendor terms respected; public pages opt-in only                                      |

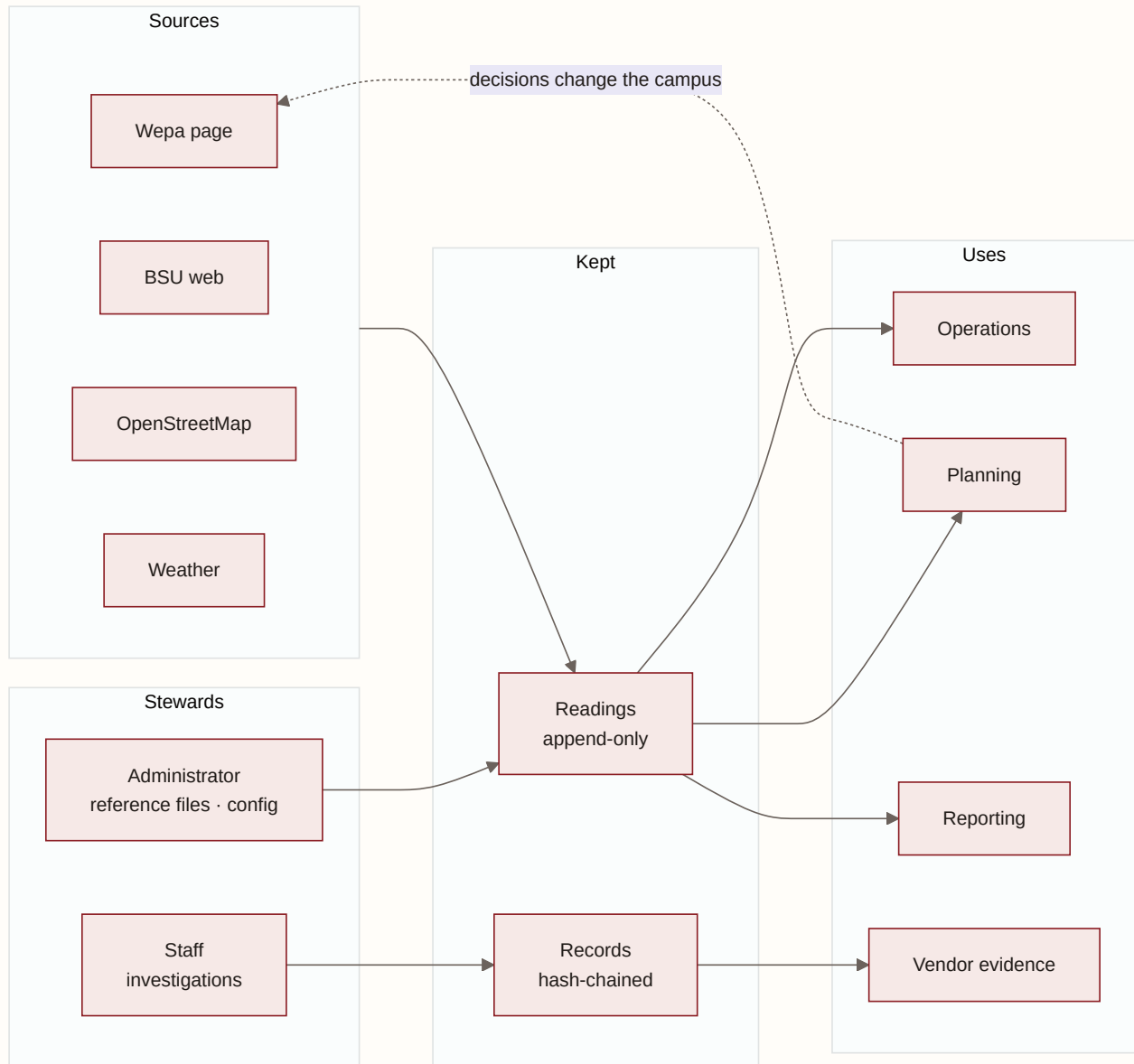
## 9.3 ASA Ethical Guidelines for Statistical Practice

| GUIDELINE AREA                            | IN PRACTICE                                                                                                                        |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Professional integrity and accountability | Methods named and documented; sample sizes on every statistic; uncertainty shown as intervals                                      |
| Integrity of data and methods             | Raw data never edited; censoring handled (Kaplan–Meier); confounding controlled (Mantel–Haenszel); models gated against a baseline |
| Responsibilities to stakeholders          | Plain-language summaries; "not enough evidence" said plainly; correlation not presented as cause                                   |
| Responsibilities to data subjects         | Minimal personal data; staff directory visibility limited by role                                                                  |

## 9.4 DASCA role families

| ROLE FAMILY      | WORK DEMONSTRATED IN THIS PRODUCT                                                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data engineering | Resilient once-a-minute ingestion; append-only storage with Parquet compaction; incremental loading; imports and de-duplication; backup collection with signed transfer            |
| Data analysis    | KPI definitions with gates; dashboards; stories; exports; executive reporting                                                                                                      |
| Data science     | Survival analysis, SPC, robust regression, empirical Bayes, Monte Carlo, stratified rate ratios, gated machine learning with walk-forward evaluation, an evidence-bound AI analyst |

## 9.5 Data ecology



**Figure 9.1 · The data ecosystem.** Who supplies, who stewards, what is kept and how it is used. Planning decisions (a new printer, longer desk hours) change the very system the data describes, which is why history is kept raw and recomputed.

## 9.6 Documentation and standards recommendations

| #  | RECOMMENDATION                                                           | WHY                                                                               | EFFORT                      |
|----|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------|
| 1  | Single sign-on through BSU's Microsoft Entra ID with MFA                 | Closes the largest security gap; ties actions to people                           | Medium, needs BSU IT        |
| 2  | Formal WCAG 2.2 AA audit with assistive technology                       | Accessibility claims should be tested, not assumed                                | Low                         |
| 3  | Architecture decision records ( docs/ad1/ )                              | Keeps the reasons behind choices (no imputation, gated ML) with the code          | Low                         |
| 4  | Software bill of materials (CycloneDX) per release                       | Supply-chain transparency; complements the vulnerability check                    | Low                         |
| 5  | Data retention policy for readings, visits and records                   | States how long each kind of data is kept and why                                 | Low, needs a decision owner |
| 6  | Model card for the outage-risk model, published with each retrain        | Documents data, metrics, limits and intended use (the app already computes these) | Low                         |
| 7  | ITSM integration by reference number (read-only link to tickets)         | Traceability without duplicating BSU's ticketing system                           | Medium                      |
| 8  | Run the container as an unprivileged user                                | Limits the damage of any future flaw                                              | Low                         |
| 9  | Written agreement with Wepa and BSU before any public use of status data | Required by Wepa's terms                                                          | Low                         |
| 10 | Docs as code: build this document in CI on each release                  | Documentation stays in step with the product                                      | Low                         |

# Reference

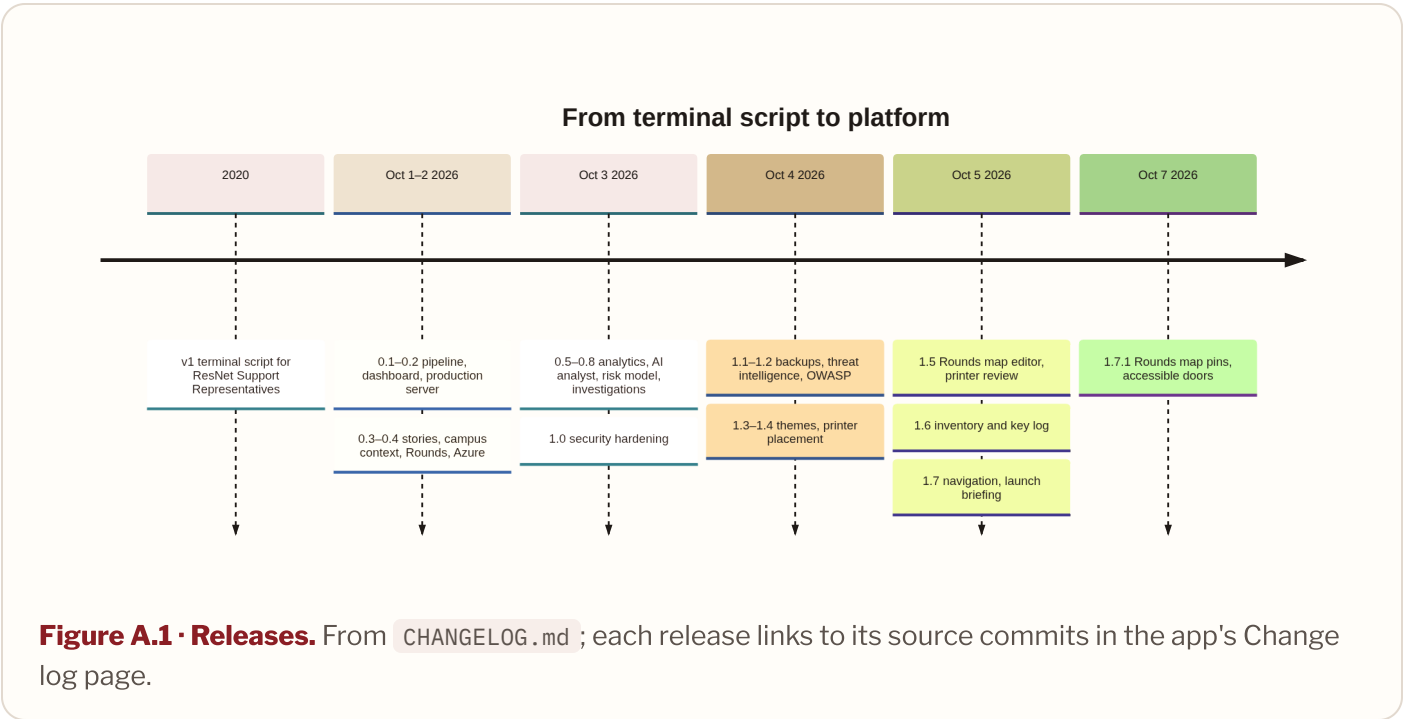
## A.1 Configuration

| SETTING                                                                     | PURPOSE                                                   |
|-----------------------------------------------------------------------------|-----------------------------------------------------------|
| WEPA_DATA_DIR , WEPA_CAMPUS_DIR                                             | Where readings and campus files are kept                  |
| WEPA_COLLECT                                                                | 1 to collect in this process, 0 for dashboard only        |
| WEPA_BASIC_AUTH , WEPA_SECRET_KEY                                           | Administrator credential (hashed) and session signing key |
| WEPA_BREACH_CHECK                                                           | 0 turns off the breached-password lookup                  |
| WEPA_PUBLIC_STATUS                                                          | 1 makes student status pages public (agreement required)  |
| WEPA_AI_PROVIDER , WEPA_AI_MODEL , ANTHROPIC_API_KEY , WEPA_AI_MAX_PER_HOUR | Optional AI analyst and its spending cap                  |
| WEPA_VULN_CHECK , WEPA_THREAT_INTEL , WEPA_NVD_API_KEY                      | Vulnerability checks and threat-intelligence look-ups     |
| WEPA_GITHUB_TOKEN , WEPA_GITHUB_ISSUES_TOKEN , WEPA_GITHUB_REPO             | Starting updates; filing feature requests                 |
| WEPA_FEEDBACK_NAMES                                                         | 1 adds the sender's name to feature-request issues        |
| WEPA_PEER_KEY                                                               | Shared key for backup collectors (32+ characters)         |
| WEPA_SUPPLY_LEAD_DAYS                                                       | Days a supply order takes (default 7)                     |

# A.2 Commands

| COMMAND                                                  | DOES                                               |
|----------------------------------------------------------|----------------------------------------------------|
| <code>python -m wepa_monitor start</code>                | Collect and run the dashboard on one computer      |
| <code>python -m wepa_monitor demo --days 120</code>      | Generate synthetic demo history                    |
| <code>python -m wepa_monitor dashboard --demo</code>     | Run the dashboard on demo data                     |
| <code>python -m wepa_monitor campus --force</code>       | Refresh calendar, halls, library hours and classes |
| <code>python -m wepa_monitor train-risk</code>           | Train and evaluate the outage-risk model now       |
| <code>python -m wepa_monitor users ...</code>            | Manage staff accounts on the running app           |
| <code>python -m wepa_monitor selfcheck [--vulns]</code>  | Run the self-check                                 |
| <code>python -m wepa_monitor backup --primary URL</code> | Stand by as a backup collector                     |
| <code>python -m wepa_monitor export --tag mac</code>     | Package local data for import elsewhere            |
| <code>python -m wepa_monitor kml</code>                  | Building pins for Google Earth                     |

# A.3 Release history



## A.4 Glossary

|                       |                                                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Availability</b>   | Share of observed time a printer could print.                                                                                                    |
| <b>Censored</b>       | An incident whose start or end is outside what was observed; kept, but handled so it doesn't bias averages.                                      |
| <b>CRISP-DM</b>       | Cross-Industry Standard Process for Data Mining: business understanding, data understanding, data preparation, modeling, evaluation, deployment. |
| <b>CTQ</b>            | Critical to quality: a measurable characteristic that matters to the customer.                                                                   |
| <b>CVE, GHSA</b>      | Public identifiers for a security vulnerability (Common Vulnerabilities and Exposures; GitHub Security Advisory).                                |
| <b>Degraded</b>       | Printing, with a warning.                                                                                                                        |
| <b>Desk hours</b>     | When a support team's desk is staffed.                                                                                                           |
| <b>EPSS</b>           | Exploit Prediction Scoring System: probability a CVE is exploited in the next 30 days.                                                           |
| <b>KEV</b>            | CISA's Known Exploited Vulnerabilities catalog.                                                                                                  |
| <b>MTBF, MTTR</b>     | Mean time between failures; mean time to repair.                                                                                                 |
| <b>No signal</b>      | No reading for 5 minutes; status unknown.                                                                                                        |
| <b>Observed time</b>  | Time covered by readings, at most 5 minutes per reading.                                                                                         |
| <b>Out of service</b> | Cannot print (a red row on Wepa's page).                                                                                                         |
| <b>ResNet</b>         | BSU's residential network and technology support team, which supports the residence-hall printers.                                               |
| <b>RSR</b>            | ResNet Support Representative: a student staff member.                                                                                           |
| <b>SIPOC</b>          | Suppliers, inputs, process, outputs, customers.                                                                                                  |

## A.5 Sources and further reading

- Chapman, P. et al. (2000). *CRISP-DM 1.0: Step-by-step data mining guide*.
- ISO/IEC 25010, ISO/IEC/IEEE 26514, ISO/IEC/IEEE 15289, ISO 8601, ISO/IEC 27001.
- Object Management Group: BPMN 2.0, CMMN 1.1, DMN.
- OWASP Top 10:2025, [owasp.org/Top10/2025](https://owasp.org/Top10/2025).
- NIST Special Publication 800-63B, Digital Identity Guidelines.

- ACM Code of Ethics and Professional Conduct (2018).
- American Statistical Association, Ethical Guidelines for Statistical Practice.
- Data Science Council of America (DASCA), credential frameworks.
- Kaplan, E. L. and Meier, P. (1958); Theil, H. (1950) and Sen, P. K. (1968); Mantel, N. and Haenszel, W. (1959).
- OpenStreetMap contributors (ODbL); Open-Meteo (CC BY 4.0); OSV.dev; CISA KEV; NIST NVD; Microsoft MSRC; FIRST EPSS; Exploit-DB; Metasploit Framework.

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**BSU Student Printing Ops** · product documentation SPO-DOC-001 · version 1.7.0 · October 2026.

Designed, built and documented by **Nicholas Freberg**, Bridgewater State University Class of 2020 and former BSU ResNet Support Representative (2018–2020), for the students and staff who keep campus printing.

Set in Libre Franklin, Source Serif 4 and IBM Plex Mono, in BSU crimson on campus stone. An independent portfolio project; not an official publication of Bridgewater State University or Wepa. Product names belong to their owners.